Topics in Mathematics Science VII

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Introduction to Koszul Duality

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Convention

Throughout the course, $\Bbbk$ is a field. All rings are unital and associative. All modules are right modules unless otherwise specified. When we discuss dimensions, we mean dimensions as $\Bbbk$-vector spaces.

There are a lot of contents not covered in lecture:

> This is supplementary material; often for 'unnecessarily long pedantic check of definitions' or content of similar nature – things that are only there for rigour

> This is advanced material; things that I personally would like to add, from personal experience in the theory, but definitely not necessary for lecture purpose.

Neither of these are rigourously checked since they are mostly produced by AI, so the reader should read these contents with extra cares themselves. All other contents are drafted by AI and QA'ed (heavily edited) by myself.

1 Graded algebras, modules, and duals

1.1 Algebras and modules

Definition 1.1. *A $\Bbbk$-algebra is a $\Bbbk$-vector space with an associative unital multiplication that is bilinear, meaning linear in each variable.*

A right A-module is understood to be a $\Bbbk$-vector space M with a bilinear action satisfying $(ma)b = m(ab)$ and $m1_A = m$ for $m \in M$ and $a, b \in A$. A module homomorphism $f : M \to N$ is linear and

satisfies $f(ma) = f(m)a$. A homomorphism of $\Bbbk$-algebras is a linear map preserving multiplication and the identity.

Example 1.2. *(1) The matrix ring* $\mathrm{Mat}_n(\Bbbk)$ *is a* $\Bbbk$*-algebra. The space of* $1 \times n$ *row vectors is a right* $\mathrm{Mat}_n(\Bbbk)$*-module by matrix multiplication.*

(2) Let $\Bbbk A_n$ *be the upper triangular matrix ring, which is also a* $\Bbbk$*-algebra.*

$$\Bbbk A_n = \begin{pmatrix} \Bbbk & \Bbbk & \cdots & \Bbbk \\ 0 & \Bbbk & \cdots & \Bbbk \\ 0 & 0 & \ddots & \vdots \\ 0 & \cdots & 0 & \Bbbk \end{pmatrix}$$

For those familiar with path algebras, $\Bbbk A_n$ *is the path algebra of a linearly oriented* A_n *quiver, which explains the notation. We use matrix calculations, so no knowledge of quivers is required. Write* e_{ij} *for the matrix with* 1 *in entry* (i, j) *and zero elsewhere. Our convention composes paths from left to right:* e_{ij} *represents* $i \to j$*, and* $e_{ij}e_{j\ell} = e_{i\ell}$*.*

1.2 Categories and functors

We organise modules together with their homomorphisms. A category records both: specifying the objects alone does not specify a category.

Definition 1.3. *A category* $\mathcal{C}$ *consists of objects, a set* $\mathrm{Hom}_{\mathcal{C}}(X, Y)$ *of morphisms* $f : X \to Y$ *for each pair of objects, and composition sending* $f : X \to Y$ *and* $g : Y \to Z$ *to* $g \circ f : X \to Z$*. Each object* X *has an identity* $\mathrm{id}_X : X \to X$*, and*

$$h \circ (g \circ f) = (h \circ g) \circ f, \qquad \mathrm{id}_Y \circ f = f = f \circ \mathrm{id}_X \quad \text{for all composable } f, g, h.$$

A morphism $f : X \to Y$ is an *isomorphism* if some $g : Y \to X$ satisfies $g \circ f = \mathrm{id}_X$ and $f \circ g = \mathrm{id}_Y$; we then write $X \cong Y$. A *full subcategory* of $\mathcal{C}$ consists of chosen objects of $\mathcal{C}$, with all morphisms between them and the same composition.

Example 1.4. *For a* $\Bbbk$*-algebra* A*, we use the notation*

$$\begin{aligned} \mathrm{Mod}\, A &:= \textit{category of right } A\textit{-modules and } A\textit{-linear maps}, \\ \mathrm{mod}\, A &:= \textit{full subcategory of finitely generated right } A\textit{-modules}. \end{aligned}$$

In either category, $\mathrm{Hom}_A(M, N)$ *is the space of* A*-linear maps* $M \to N$*; composition is the usual composition of maps. For* $A = \Bbbk$*, these are the categories of all and of finite-dimensional vector spaces, respectively.*

Note that, if $\dim_{\Bbbk} A < \infty$*, then* $M \in \mathrm{mod}\, A \Leftrightarrow M$ *is finite-dimensional (*$\because$ *finite generation implies that* M *is a quotient of* $A^{\oplus n}$ *some* n*, and a finite* $\Bbbk$*-basis of* M *is clearly a finite generating set.)*

Example 1.5. *For* $M \in \mathrm{mod}\, A$*, let* $\mathrm{add}\, M := \mathrm{add}_{\mathrm{mod}\, A}(M)$ *be the full subcategory with*

$$N \in \mathrm{add}\, M \quad \Longleftrightarrow \quad N \oplus N' \cong M^{\oplus r} \quad \text{for some } N' \in \mathrm{mod}\, A \text{ and integer } r \geq 0.$$

Thus $\mathrm{add}\, M$ *consists of direct summands of finite direct sums of* M*.*

In modern representation theory, remembering A and $\mathrm{add}\, A$ holds the same categorical information is crucial in utilising cateogircal technique in representation theory; this allows one to push many results to a more abstract framework.

Definition 1.6. *A functor $F : \mathcal{C} \to \mathcal{D}$ assigns an object $F(X)$ to each object X and a morphism $F(f) : F(X) \to F(Y)$ to each $f : X \to Y$, such that*

$$F(\mathrm{id}_X) = \mathrm{id}_{F(X)}, \qquad F(g \circ f) = F(g) \circ F(f) \quad \text{for all composable } f, g.$$

For example, forgetting the A-action gives a functor $\operatorname{Mod} A \to \operatorname{Mod} \Bbbk$: it keeps the underlying vector spaces and linear maps. The *identity functor* $\mathrm{id}_{\mathcal{C}}$ leaves every object and morphism unchanged; functors compose by applying one after the other. Every functor preserves isomorphisms, since $F(f^{-1})$ is inverse to $F(f)$.

The *opposite category* $\mathcal{C}^{\mathrm{op}}$ has the same objects as $\mathcal{C}$, with $\operatorname{Hom}_{\mathcal{C}^{\mathrm{op}}}(X, Y) = \operatorname{Hom}_{\mathcal{C}}(Y, X)$ and composition reversed. A *contravariant functor* from $\mathcal{C}$ to $\mathcal{D}$ means a functor $\mathcal{C}^{\mathrm{op}} \to \mathcal{D}$. Thus it sends $f : X \to Y$ in $\mathcal{C}$ to $F(f) : F(Y) \to F(X)$ and satisfies $F(g \circ f) = F(f) \circ F(g)$. Duals will provide our main examples.

Definition 1.7. *For functors $F, G : \mathcal{C} \to \mathcal{D}$, a natural transformation $\eta : F \to G$ is a family of morphisms $\eta_X : F(X) \to G(X)$ such that*

$$\begin{array}{ccc} F(X) & \xrightarrow{\eta_X} & G(X) \\ {\scriptstyle F(f)}\downarrow & & \downarrow{\scriptstyle G(f)} \\ F(Y) & \xrightarrow[\eta_Y]{} & G(Y) \end{array} \qquad G(f) \circ \eta_X = \eta_Y \circ F(f) \quad \text{for every } f : X \to Y. \tag{1.1}$$

If every η_X is an isomorphism, η is a natural isomorphism, written $F \cong G$.

The square in (1.1) is said to *commute*: its two routes from $F(X)$ to $G(Y)$ give the same morphism. For example, for fixed $c \in \Bbbk$, the maps $c \,\mathrm{id}_M : M \to M$ define a natural transformation $\mathrm{id}_{\operatorname{Mod} A} \to \mathrm{id}_{\operatorname{Mod} A}$ because $f(cm) = cf(m)$ for every module homomorphism $f : M \to N$. It is a natural isomorphism when $c \neq 0$.

A functor $F : \mathcal{C} \to \mathcal{D}$ is an *equivalence* if there is a functor $G : \mathcal{D} \to \mathcal{C}$ with natural isomorphisms $GF \cong \mathrm{id}_{\mathcal{C}}$ and $FG \cong \mathrm{id}_{\mathcal{D}}$. A *duality* between $\mathcal{C}$ and $\mathcal{D}$ is an equivalence $\mathcal{C}^{\mathrm{op}} \to \mathcal{D}$. Thus an equivalence recovers objects and maps up to natural isomorphism; a duality also reverses the direction of maps.

Definition 1.8. *Let A be a $\Bbbk$-algebra, and let $L, M, N \in \operatorname{Mod} A$, with $i : L \to M$ and $p : M \to N$ module homomorphisms. The sequence $L \xrightarrow{i} M \xrightarrow{p} N$ is exact at M if $\operatorname{Im} i = \operatorname{Ker} p$. A sequence of module homomorphisms is exact if it is exact at every term with an incoming and an outgoing map. A short exact sequence is an exact sequence*

$$0 \longrightarrow L \xrightarrow{i} M \xrightarrow{p} N \longrightarrow 0, \qquad \operatorname{Ker} i = 0, \quad \operatorname{Im} i = \operatorname{Ker} p, \quad \operatorname{Im} p = N. \tag{1.2}$$

Definition 1.9. *Let A, B be $\Bbbk$-algebras. A functor $F : \operatorname{Mod} A \to \operatorname{Mod} B$ is exact if $F(0) = 0$ and every short exact sequence (1.2) gives a short exact sequence*

$$0 \longrightarrow F(L) \xrightarrow{F(i)} F(M) \xrightarrow{F(p)} F(N) \longrightarrow 0.$$

An exact functor preserves injective and surjective maps: apply it to $0 \to L \xrightarrow{i} M \to M/i(L) \to 0$ and $0 \to \operatorname{Ker} p \to M \xrightarrow{p} N \to 0$, respectively. For a functor $F : (\operatorname{Mod} A)^{\mathrm{op}} \to \operatorname{Mod} B$, exactness instead means that (1.2) gives

$$0 \longrightarrow F(N) \xrightarrow{F(p)} F(M) \xrightarrow{F(i)} F(L) \longrightarrow 0.$$

Thus an exact contravariant functor exchanges injective and surjective maps.

Example 1.10 (𝕜-linear duality). *For a vector space V, define its 𝕜-linear dual by $D(V) := \mathrm{Hom}_{\Bbbk}(V, \Bbbk)$. For a linear map $u : V \to W$, set*

$$D(u) : D(W) \longrightarrow D(V), \qquad \lambda \longmapsto \lambda \circ u.$$

The identities $D(\mathrm{id}_V) = \mathrm{id}_{D(V)}$ and $D(v \circ u) = D(u) \circ D(v)$ make $D : (\operatorname{Mod} \Bbbk)^{\mathrm{op}} \to \operatorname{Mod} \Bbbk$ a functor. The functor D is exact.

> *For a short exact sequence (1.2) of vector spaces, $D(p)$ is injective because p is surjective, and $D(i)$ is surjective because a functional on a subspace extends to the whole vector space. If $\lambda \in D(M)$ satisfies $\lambda \circ i = 0$, then*
>
> $$\mu : N \longrightarrow \Bbbk, \qquad p(m) \longmapsto \lambda(m)$$
>
> *is well-defined since $\operatorname{Ker} p = i(L)$, and $\lambda = \mu \circ p = D(p)(\mu)$. Conversely, $p \circ i = 0$ gives $D(i) \circ D(p) = 0$, so $\operatorname{Ker} D(i) = \operatorname{Im} D(p)$. Thus D exchanges injective and surjective linear maps.*

For every vector space V, evaluation defines a natural injective map

$$\mathrm{ev}_V : V \longrightarrow D(D(V)), \qquad v \longmapsto (\lambda \mapsto \lambda(v)). \tag{1.3}$$

If V is finite-dimensional, choose a basis $v_1, \dots, v_n$ and its dual basis $v_1^*, \dots, v_n^*$, defined by $v_i^*(v_j) = \delta_{ij}$. Then

$$\mathrm{ev}_V^{-1} : D(D(V)) \longrightarrow V, \qquad \Phi \longmapsto \sum_{i=1}^{n} \Phi(v_i^*) v_i.$$

Consequently, $D : (\operatorname{mod} \Bbbk)^{\mathrm{op}} \to \operatorname{mod} \Bbbk$ is a duality.

> *If $v \neq 0$, choose $\lambda \in D(V)$ with $\lambda(v) \neq 0$; hence ev_V is injective. For $u : V \to W$, $v \in V$, and $\lambda \in D(W)$,*
>
> $$D(D(u))(\mathrm{ev}_V(v))(\lambda) = \lambda(u(v)) = \mathrm{ev}_W(u(v))(\lambda),$$
>
> *which proves naturality.*
>
> *In finite dimension, writing $\lambda = \sum_{i=1}^n \lambda(v_i) v_i^*$ gives $\Phi(\lambda) = \lambda(\sum_{i=1}^n \Phi(v_i^*) v_i)$. Thus the displayed formula is the inverse of (1.3), and is independent of the chosen basis.*

1.3 Semisimple algebras

Definition 1.11. *A module L is simple if $L \neq 0$ and its only submodules are 0 and L. A module is semisimple if it is a direct sum of simple modules. An algebra B is semisimple if it is semisimple as a right module B_B over itself. The Jacobson radical $\operatorname{rad} B$ is the intersection of all maximal right ideals of B.*

We use the following standard ring-theoretic facts without proof. Recall that a *division 𝕜-algebra* is a non-zero 𝕜-algebra whose non-zero elements are invertible.

Theorem 1.12 (Artin–Wedderburn). *For a non-zero finite-dimensional 𝕜-algebra B,*

$$B \text{ is semisimple} \iff \operatorname{rad} B = 0 \iff B \cong \prod_{i=1}^{r} \mathrm{Mat}_{n_i}(D_i),$$

where $r, n_i \geq 1$ and each D_i is a finite-dimensional division 𝕜-algebra. If 𝕜 is algebraically closed, then $D_i = \Bbbk$ for every i. For semisimple B, every left or right B-module is semisimple, and every submodule has a complement. In particular, the definition is independent of the chosen side.

Corollary 1.13. *Every finite-dimensional left or right module over a semisimple algebra S is a direct summand of $S^{\oplus n}$ for some n.*

Proof A finite generating set gives a surjection $S^{\oplus n} \twoheadrightarrow M$. By Theorem 1.12, its kernel has a complement, which maps isomorphically onto M. □

Example 1.14. *The algebras $\Bbbk^{\oplus r}$ and $\mathrm{Mat}_n(\Bbbk)$ are semisimple by Theorem 1.12. Upper triangular matrices need not form a semisimple algebra: for $A = \Bbbk A_2$,*

$$\operatorname{rad} A = \Bbbk e_{12} \neq 0, \qquad A/\operatorname{rad} A \cong \Bbbk^{\oplus 2}.$$

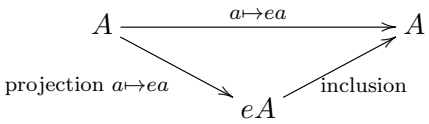
> *Put $J = \Bbbk e_{12}$, so $J^2 = 0$. For a simple right module L, simplicity gives $LJ = 0$ or $LJ = L$; the second alternative implies $L = LJ^2 = 0$, a contradiction. Hence J is contained in every maximal right ideal. The quotient $A/J \cong \Bbbk^{\oplus 2}$ has exactly two maximal right ideals, whose inverse images have intersection*
>
> $$(\Bbbk e_{11} \oplus \Bbbk e_{12}) \cap (\Bbbk e_{12} \oplus \Bbbk e_{22}) = J.$$

Definition 1.15. *An element $e \in A$ is idempotent if $e^2 = e$. Idempotents e, f are orthogonal if $ef = fe = 0$. A non-zero idempotent is primitive if it is not the sum of two non-zero orthogonal idempotents. A finite family $e_1, \ldots, e_r$ of pairwise orthogonal idempotents is complete if $\sum_{i=1}^r e_i = 1_A$.*

Remark 1.16. For orthogonal idempotents $e, f \in A$,

$$(e + f)A = eA \oplus fA.$$

In particular, we always have $A \cong eA \oplus (1 - e)A$. This strengthen the viewpoint of idempotent-as-local-identity: left-multiplying e defines a canonical projection onto the direct summand eA of A, and yields a commutative diagram



So multiplying e means we work inside the direct summand eA. The corner subalgebra $eAe = \{eae \mid a \in A\}$ has identity e (note that this is not a unital subalgera; but rather a 'subcategory'). For $A = \prod_{i=1}^r A_i$, the coordinate idempotents $e_i = (0, \ldots, 1_{A_i}, \ldots, 0)$ are a complete orthogonal family and are the identities of the individual factors. In $A = \Bbbk A_n$, the diagonal matrix units $e_{11}, \ldots, e_{nn}$ are primitive, since $e_{ii}Ae_{ii} = \Bbbk e_{ii}$. For $n \geq 2$, the sum $e = e_{11} + \cdots + e_{n-1,n-1}$ gives the corner

$$eAe = \left\{ \begin{pmatrix} B & 0 \\ 0 & 0 \end{pmatrix} \;\middle|\; B \in \Bbbk A_{n-1} \right\} \cong \Bbbk A_{n-1}.$$

Idempotents also describe maps out of the summands of A_A.

Lemma 1.17 (Yoneda lemma). *For an idempotent $e \in A$ and a right A-module M, evaluation at e gives a natural vector space isomorphism*

$$\operatorname{Hom}_A(eA, M) \xrightarrow{\sim} Me, \qquad f \longmapsto f(e), \quad \text{with inverse} \quad m \longmapsto (ea \mapsto ma).$$

Proof We have $f(e) = f(e)e$ and $f(ea) = f(e)a$. If $m = me$, the map $ea \mapsto ma$ is well-defined because $ea = ea'$ implies $ma = mea = mea' = ma'$. These formulas prove that the maps are inverse; postcomposition by a module map $M \to N$ proves naturality. □

1.4 Graded algebras

Definition 1.18. *Let G be a group with identity e. A G-graded $\Bbbk$-algebra is a $\Bbbk$-algebra A with $1_A \in A_e$ and a decomposition*

$$A = \bigoplus_{g \in G} A_g \quad \text{such that} \quad A_g A_h \subset A_{gh} \quad \text{for all } g, h \in G.$$

An element of A_g is homogeneous of degree g.

Convention. *Unless stated otherwise, a graded algebra is $\mathbb{Z}$-graded and concentrated in non-negative degrees.*

Note that the decomposition is into a direct sum, so every element of a graded algebra A can always be written as a *finite* sum of homogeneous elements.

Example 1.19. *For the upper triangular matrix ring $\Bbbk A_n$ of Example 1.2, the decomposition*

$$(\Bbbk A_n)_d = \operatorname{span}_\Bbbk\{e_{ij} \mid j - i = d\}$$

makes $\Bbbk A_n$ a graded algebra.

Example 1.20. *The algebra $\Bbbk A_3$ also has a $\mathbb{Z}^{\oplus 2}$-grading with*

$$\deg e_{ii} = (0,0), \qquad \deg e_{12} = (1,0), \qquad \deg e_{23} = (0,1), \qquad \deg e_{13} = (1,1).$$

The homomorphism $\mathbb{Z}^{\oplus 2} \to \mathbb{Z}$, $(p,q) \mapsto p+q$, gives the grading of Example 1.19. Alternatively, setting $\deg e_{ij} = 2(j-i)$ gives a $\mathbb{Z}$-grading with $A_1 = 0$ but $A_2 \neq 0$.

Definition 1.21. *A homomorphism of graded algebras $f : A \to B$ is a $\Bbbk$-algebra homomorphism with $f(A_i) \subset B_i$ for all $i \in \mathbb{Z}$.*

Definition 1.22. *A graded algebra A is locally finite if $\dim_\Bbbk A_i < \infty$ for every $i \in \mathbb{Z}$.*

1.5 Graded modules

Definition 1.23. *Let $A = \bigoplus_{g \in G} A_g$ be a G-graded algebra. A G-graded right A-module is a right A-module M with a direct sum decomposition*

$$M = \bigoplus_{g \in G} M_g \quad \text{such that} \quad M_g A_h \subset M_{gh} \quad \text{for all } g, h \in G.$$

An element of M_g is called homogeneous of degree g.

A G-graded A-module $M = \bigoplus_{g \in G} M_g$ is locally finite if $\dim_\Bbbk M_g < \infty$ for all $g \in G$.

Unless specified otherwise, graded modules are $\mathbb{Z}$-graded; their degrees need not be non-negative.

For an algebra A, the opposite algebra A^{op} has the same vector space as A, with multiplication $a^{\mathrm{op}} b^{\mathrm{op}} = (ba)^{\mathrm{op}}$. A left A-module is a right A^{op}-module by $m \cdot a^{\mathrm{op}} = am$. Thus $\operatorname{Mod}(A^{\mathrm{op}})$ is the category of left A-modules and left A-linear maps; not to be confused with the opposite category $(\operatorname{Mod} A)^{\mathrm{op}}$ though. For a $\mathbb{Z}$-graded algebra $A = \bigoplus_{i \geq 0} A_i$,

$$\text{the opposite algebra is also graded with } (A^{\mathrm{op}})_i = A_i.$$

Definition 1.24. *Let M be a graded A-module and let $d \in \mathbb{Z}$. The degree shift, or grading shift, $M(d)$ is the graded A-module whose underlying A-module is M, but whose grading is*

$$M(d)_j = M_{j+d}.$$

By Definition 1.24, $1 \in A_0 = A(-d)_d$, so $A(-d)$ has a generator in degree d. If M is concentrated in degree 0, then $M(d)$ is concentrated in degree $-d$.

Example 1.25. *For $A = \Bbbk A_3$ with the grading of Example 1.19, the module $P_1 := e_{11}A$ consists of matrices supported on the first row. Its basis e_{11}, e_{12}, e_{13} has degrees $0, 1, 2$, respectively; in $P_1(-2)$, the same basis has degrees $2, 3, 4$.*

Definition 1.26. *Let M, N be graded A-modules. A homomorphism $f : M \to N$ is homogeneous of degree d if $f(M_j) \subset N_{j+d}$ for all $j \in \mathbb{Z}$. A homomorphism of graded modules means a homogeneous A-module homomorphism of degree 0, also said to preserve degrees.*

Example 1.27. *For $A = \Bbbk A_3$, left multiplication by e_{12} yields a map*

$$\begin{aligned} e_{22}A &\longrightarrow e_{11}A, \\ \begin{pmatrix} 0 & 0 & 0 \\ 0 & u & v \\ 0 & 0 & 0 \end{pmatrix} &\longmapsto \begin{pmatrix} 0 & u & v \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \qquad (u, v \in \Bbbk). \end{aligned}$$

The u and v entries have degrees $0, 1$ in the source and $1, 2$ in the target, and so the map has degree 1.

Notation. *For graded right A-modules M, N, the space of homomorphisms preserving degrees is*

$$\mathrm{Hom}_A^{\mathbb{Z}}(M, N) = \{f \in \mathrm{Hom}_A(M, N) \mid f(M_j) \subset N_j \text{ for all } j \in \mathbb{Z}\}. \tag{1.4}$$

Thus $\mathrm{Hom}_A^{\mathbb{Z}}(M, N(d))$ consists of homomorphisms $M \to N$ of degree d, since $N(d)_j = N_{j+d}$. Note that, like the grading shift functor, different literature has different notation for this (there does not seem to be any widely agreed notation).

We write

$$\begin{aligned} \mathrm{GrMod}\, A &:= \text{category of graded right } A\text{-modules with morphism spaces } \mathrm{Hom}_A^{\mathbb{Z}}(M, N), \\ \mathrm{grmod}\, A &:= \text{full subcategory of } \mathrm{GrMod}\, A \text{ consisting of locally finite modules.} \end{aligned}$$

Thus $\mathrm{GrMod}(A^{\mathrm{op}})$ is the category of graded left A-modules and degree-preserving maps. Forgetting the grading gives a functor $\mathrm{GrMod}\, A \to \mathrm{Mod}\, A$.

Remark 1.28. $\mathrm{grmod}\, A$ is somewhat analogous to $\mathrm{mod}\, A$ but not really the same: If A is locally finite, every finitely generated graded A-module M is locally finite. ($\because$ a finite homogeneous generating set of a graded A-module M gives a surjection from a finite direct sum of shifts of A to M, so every M_i has to be finite-dimensional.) However, a locally finite M need not be finitely generated.

For $d \in \mathbb{Z}$, the shift of Definition 1.24 is a functor

$$(d) : \mathrm{GrMod}\, A \to \mathrm{GrMod}\, A.$$

It sends $f : M \to N$ to the same underlying map $f(d) : M(d) \to N(d)$. Since $M(d)(-d) = M$ and $f(d)(-d) = f$, the shift is an equivalence with inverse $(-d)$; it restricts to an equivalence $\mathrm{grmod}\, A \to \mathrm{grmod}\, A$.

Remark 1.29. The notion of exactness makes sense for graded module categories – a sequence in $\mathrm{GrMod}\, A$ is exact if the underlying sequence in $\mathrm{Mod}\, A$ is exact. In particular, we talk about functors invovle such categories being exact.

For $L, M, N \in \operatorname{GrMod} A$ with maps $i : L \to M$ and $p : M \to N$, the sequence $0 \to L \xrightarrow{i} M \xrightarrow{p} N \to 0$ is exact if and only if

$$0 \longrightarrow L_j \xrightarrow{i|_{L_j}} M_j \xrightarrow{p|_{M_j}} N_j \longrightarrow 0 \quad \text{is exact for every } j \in \mathbb{Z}.$$

Indeed, i and p preserve the direct sum decompositions, so their kernels and images decompose into the kernels and images of the restrictions.

Example 1.30. *Let $A = \Bbbk A_2$. Consider $P_1 := e_{11}A$ which has $\operatorname{rad} P_1 = \Bbbk e_{12}$ and $S_1 := P_1/\operatorname{rad} P_1$. Then we have a short exact sequence in* $\operatorname{GrMod} A$*:*

$$0 \longrightarrow \operatorname{rad} P_1 \xrightarrow{i} P_1 \xrightarrow{p} P_1/\operatorname{rad} P_1 \longrightarrow 0, \qquad i(ce_{12}) = ce_{12}, \quad p(m) = \overline{m}.$$

Note that $\deg \operatorname{rad} P_1 = 1$ and $\deg S_1 = 0$. Let $S_2 := \Bbbk e_{22} = e_{22}A$, which we treat as a graded A-module concentrated in degree 0. Note that $S_2 \cong \operatorname{rad} P_1$ as ungraded modules but we only have $S_2 \cong (\operatorname{rad} P_1)(1)$, which yields the following short exact sequence in $\operatorname{GrMod} A$*:*

$$0 \longrightarrow S_2(-1) \to P_1 \to P_1/\operatorname{rad} P_1 \longrightarrow 0,$$

Lemma 1.31. *For graded right A-modules M, N, there is an inclusion*

$$\bigoplus_{d\in\mathbb{Z}} \operatorname{Hom}_A^{\mathbb{Z}}(M, N(d)) \subset \operatorname{Hom}_A(M, N), \text{ which is equality when } M \in \operatorname{mod} A.$$

Proof For $f \in \operatorname{Hom}_A(M, N)$, define $f_d(m) = f(m)_{j+d}$ when $m \in M_j$, and extend linearly. For $a \in A_i$, we have $f_d(ma) = f(ma)_{j+i+d} = f(m)_{j+d}a$, so f_d is A-linear of degree d. The sum of the homogeneous Hom spaces is direct: on M_j, distinct degrees have values in distinct summands of N.

If M is finitely generated, the homogeneous components of a finite generating set form a finite homogeneous generating set. Only finitely many f_d are non-zero on these homogeneous generators, hence on M, giving $f = \sum_d f_d$. □

1.6 Homogeneous ideals and quotients

Definition 1.32. *Let $A = \bigoplus_{i\in\mathbb{Z}} A_i$ be a graded algebra. A right ideal $I \subset A$ is homogeneous if*

$$I = \bigoplus_{i\in\mathbb{Z}} (I \cap A_i).$$

Equivalently, whenever $a \in I$ and $a = a_0 + \cdots + a_N$ is its decomposition into homogeneous components, each a_i also belongs to I.

For quotient algebras we need two-sided ideals. If I is a homogeneous two-sided ideal, then

$$A/I = \bigoplus_{i\in\mathbb{Z}} A_i/(I \cap A_i)$$

is again a graded algebra.

Example 1.33. *In $A = \Bbbk A_3$ with the grading of Example 1.19, the two-sided ideal $I = (e_{13}) = \Bbbk e_{13}$ is homogeneous. Writing $\overline{e}_{ij}$ for the image of e_{ij},*

$$(A/I)_0 \cong \Bbbk^{\oplus 3}, \qquad (A/I)_1 = \Bbbk\overline{e}_{12} \oplus \Bbbk\overline{e}_{23}, \qquad (A/I)_d = 0 \quad (d \geq 2).$$

In particular, $\overline{e}_{12}\overline{e}_{23} = 0$.

Example 1.34. *Let $A = \Bbbk[x]$ with $\deg x = 1$, and let $I = (x - x^2)$. The ideal I is two-sided because A is commutative. Every element of I vanishes at $x = 1$, whereas neither x nor x^2 does. Thus $x - x^2 \in I$ but $x, x^2 \notin I$, so I is not homogeneous. Writing $\overline{f} = f + I$, we have $0 \neq \overline{x} = \overline{x^2}$. The images of A_1 and A_2 therefore intersect non-trivially, so A/I does not inherit the quotient grading. As an ungraded algebra, the quotient is semisimple:*

$$\Bbbk[x]/(x - x^2) \xrightarrow{\sim} \Bbbk \oplus \Bbbk, \qquad \overline{f} \longmapsto (f(0), f(1)).$$

The inverse sends (a, b) to $a + (b - a)\overline{x}$.

Definition 1.35. *The positive degree ideal of a graded algebra A is $A_+ := \bigoplus_{i>0} A_i$.*[1]

The subspace A_+ is a homogeneous two-sided ideal. The quotient map

$$\pi : A \longrightarrow A/A_+ \cong A_0, \qquad \sum_{i \in \mathbb{Z}} a_i \mapsto a_0, \tag{1.5}$$

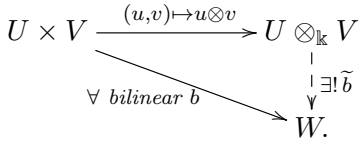
makes ${}_A(A_0)_A$ a graded bimodule concentrated in degree 0, with $a \cdot s \cdot b = \pi(a)s\pi(b)$ for $a, b \in A$ and $s \in A_0$.

1.7 Tensor products of vector spaces

Definition 1.36. *Choose a basis $\{u_i \mid i \in I\}$ of U and a basis $\{v_j \mid j \in J\}$ of V. The tensor product $U \otimes_\Bbbk V$ is the vector space with basis $\{u_i \otimes v_j \mid i \in I,\ j \in J\}$.*

The assignment $(u_i, v_j) \mapsto u_i \otimes v_j$ extends uniquely to a bilinear map $U \times V \to U \otimes_\Bbbk V$ sending $(u, v) \mapsto u \otimes v$.

Proposition 1.37 (Universal property). *Every bilinear map $b : U \times V \to W$ has a unique linear factorisation (commutative diagram)*



Consequently, $U \otimes_\Bbbk V$ is independent of the chosen bases up to a unique isomorphism preserving the bilinear map.

Proof The formula $\widetilde{b}(u_i \otimes v_j) = b(u_i, v_j)$ determines a unique linear map $\widetilde{b} : U \otimes_\Bbbk V \to W$. Bilinearity gives $\widetilde{b}(u \otimes v) = b(u, v)$ for all $u \in U$ and $v \in V$.

Suppose $t : U \times V \to T$ and $t' : U \times V \to T'$ both satisfy the universal property. They give unique linear maps $a : T \to T'$ and $b : T' \to T$ with $a \circ t = t'$ and $b \circ t' = t$. Both $b \circ a : T \to T$ and $\mathrm{id}_T : T \to T$ factor t through itself, since

$$(b \circ a) \circ t = b \circ t' = t = \mathrm{id}_T \circ t.$$

Uniqueness for (T, t) gives $b \circ a = \mathrm{id}_T$. Likewise, $(a \circ b) \circ t' = a \circ t = t' = \mathrm{id}_{T'} \circ t'$, so uniqueness for (T', t') gives $a \circ b = \mathrm{id}_{T'}$. □

[1] It is also called the augmentation ideal; we avoid that terminology.

For finite bases with $|I| = r$ and $|J| = s$, we obtain $\dim_{\Bbbk}(U \otimes_{\Bbbk} V) = rs$. We use the canonical identifications

$$(U \otimes_{\Bbbk} V) \otimes_{\Bbbk} W \xrightarrow{\sim} U \otimes_{\Bbbk} (V \otimes_{\Bbbk} W), \quad (u \otimes v) \otimes w \longmapsto u \otimes (v \otimes w),$$
$$\Bbbk \otimes_{\Bbbk} U \xrightarrow{\sim} U, \qquad\qquad c \otimes u \longmapsto cu,$$

and set $U^{\otimes_{\Bbbk} 0} := \Bbbk$.

More generally, when I, J are infinite, the tensor product can be defined as a pair. The map $(u, v) \mapsto u \otimes v$ can be defined as the linear extension of

$$\Big(\sum_{i\in I} a_i u_i, \sum_{j\in J} b_j v_j\Big) \longmapsto \sum_{(i,j)\in I\times J} a_i b_j (u_i \otimes v_j) \quad \text{for finitely supported families } (a_i)_{i\in I}, (b_j)_{j\in J}.$$

Then universal property still holds for the $U \otimes_{\Bbbk} V$.

1.8 Direct sums and direct products

For an arbitrary family $(M_i)_{i\in I}$ of right A-modules, indexed by a set I, the *direct product* and *direct sum* are

$$\prod_{i\in I} M_i = \{(m_i)_{i\in I} \mid m_i \in M_i \text{ for all } i \in I\},$$
$$\bigoplus_{i\in I} M_i = \{(m_i)_{i\in I} \in \prod_{i\in I} M_i \mid m_i = 0 \text{ for all but finitely many } i\}.$$

Addition and the A-action are componentwise; the same definitions apply to vector spaces by taking $A = \Bbbk$. Thus $\bigoplus_{i\in I} M_i \subset \prod_{i\in I} M_i$, with equality for finite I. For example, $(1, 1, \ldots) \in \prod_{i\geq 0} \Bbbk$ does not belong to $\bigoplus_{i\geq 0} \Bbbk$. In a direct sum, an element is a finite sum of its components; no notion of convergence is involved. For instance, ordinary polynomials use non-negative powers, whereas Laurent polynomials allow integer powers, but both are finite sums:

$$\Bbbk[x] = \bigoplus_{n\geq 0} \Bbbk x^n, \qquad \Bbbk[x, x^{-1}] = \bigoplus_{n\in\mathbb{Z}} \Bbbk x^n.$$

The *formal power series ring* allows arbitrary coefficients:

$$\Bbbk[[x]] = \left\{ \sum_{n\geq 0} a_n x^n \;\middle|\; a_n \in \Bbbk \text{ for every } n \geq 0 \right\} \cong \prod_{n\geq 0} \Bbbk x^n \quad \text{as vector spaces.}$$

Addition is coefficientwise, and multiplication is

$$\Big(\sum_{n\geq 0} a_n x^n\Big)\Big(\sum_{n\geq 0} b_n x^n\Big) = \sum_{n\geq 0} \Big(\sum_{j=0}^{n} a_j b_{n-j}\Big) x^n.$$

Tensor products distribute over direct sums in either variable. For vector spaces U, V and arbitrary families $(U_i)_{i\in I}$, $(V_j)_{j\in J}$ indexed by sets I, J, there are natural linear isomorphisms

$$\begin{aligned} \Big(\bigoplus_{i\in I} U_i\Big) \otimes_{\Bbbk} V &\xrightarrow{\sim} \bigoplus_{i\in I} (U_i \otimes_{\Bbbk} V), \quad (u_i)_{i\in I} \otimes v \longmapsto (u_i \otimes v)_{i\in I}, \\ U \otimes_{\Bbbk} \Big(\bigoplus_{j\in J} V_j\Big) &\xrightarrow{\sim} \bigoplus_{j\in J} (U \otimes_{\Bbbk} V_j), \quad u \otimes (v_j)_{j\in J} \longmapsto (u \otimes v_j)_{j\in J}. \end{aligned} \tag{1.6}$$

The displayed bilinear assignments induce these linear maps by Proposition 1.37. On the ith summand, the inverse of the first map sends $u_i \otimes v$ to the element with u_i in coordinate i, tensored with v. On the jth summand, the inverse of the second sends $u \otimes v_j$ to u tensored with the element having v_j in coordinate j. Extending over the direct sums gives the two inverses.

For direct products, the corresponding map need not be an isomorphism.

Lemma 1.38. *For an arbitrary family $(U_i)_{i\in I}$ of vector spaces, indexed by a set I, and a vector space V, the coordinate projections define a linear map*

$$\Big(\prod_{i\in I} U_i\Big) \otimes_{\Bbbk} V \longrightarrow \prod_{i\in I}(U_i \otimes_{\Bbbk} V), \qquad (u_i)_i \otimes v \longmapsto (u_i \otimes v)_i. \tag{1.7}$$

This map is injective, and is an isomorphism if I is finite or V is finite-dimensional. The same statements hold with the two tensor factors reversed.

Proof The bilinear map $\big(\prod_{i\in I} U_i\big) \times V \to \prod_{i\in I}(U_i \otimes_{\Bbbk} V)$, $((u_i)_i, v) \mapsto (u_i \otimes v)_i$, induces (1.7) by Proposition 1.37. Every element can be written $\sum_{\ell=1}^{r} u^{(\ell)} \otimes v_\ell$ with the v_ℓ linearly independent: choose a basis of the span of its finitely many second factors. If its image is zero, then $\sum_{\ell=1}^{r} u_i^{(\ell)} \otimes v_\ell = 0$ for every i. The tensor-product basis construction gives $u_i^{(\ell)} = 0$ for all i, ℓ, proving injectivity. If $v_1, \dots, v_r$ is a basis of V, the inverse is

$$\prod_{i\in I}(U_i \otimes_{\Bbbk} V) \longrightarrow \Big(\prod_{i\in I} U_i\Big) \otimes_{\Bbbk} V, \qquad \Big(\sum_{\ell=1}^{r} u_i^{(\ell)} \otimes v_\ell\Big)_i \longmapsto \sum_{\ell=1}^{r} (u_i^{(\ell)})_i \otimes v_\ell.$$

For finite I, products equal sums. Reversing the tensor factors gives the other statements, since $U \otimes_{\Bbbk} V \to V \otimes_{\Bbbk} U$, $u \otimes v \mapsto v \otimes u$, is an isomorphism. □

Example 1.39. *The comparison in* (1.7) *need not be surjective when V is infinite-dimensional. Take $U_i = \Bbbk$ for $i \geq 1$ and $V = \bigoplus_{j\geq 1} \Bbbk v_j$. Under $\Bbbk \otimes_{\Bbbk} V \cong V$, the image of the product comparison in* (1.7) *is*

$$\left\{ (w_i)_i \in \prod_{i\geq 1} V \;\middle|\; \dim_{\Bbbk} \operatorname{span}_{\Bbbk}\{w_i \mid i \geq 1\} < \infty \right\}.$$

Indeed, the image of $\sum_{\ell=1}^{r} (c_i^{(\ell)})_i \otimes w_\ell$ has every entry in $\operatorname{span}_{\Bbbk}\{w_1, \dots, w_r\}$. Conversely, expand all entries in a basis of their common finite-dimensional span. Since the v_i are linearly independent, $(v_i)_i$ is not in the image.

1.9 Bimodules and enveloping algebras

Definition 1.40. *For $\Bbbk$-algebras B, C, a (B,C)-bimodule M is a vector space M where left B- and right C-module actions commute, i.e. $(bm)c = b(mc)$ for all $b \in B$, $c \in C$, and $m \in M$. When $B = C = A$, we call M an A-bimodule. For graded B, C, a bimodule M is graded if $M = \bigoplus_{i\in\mathbb{Z}} M_i$ as vector space and $B_i M_j C_k \subset M_{i+j+k}$ for all $i, j, k \in \mathbb{Z}$.*

Convention.

- *Instead of saying (B,C)-bimodule everytime, we usually just write ${}_B M_C$. Write ${}_B\operatorname{Mod}_C$ for the category of these bimodules and maps preserving both actions.*
- *Bilinearity implies that $\Bbbk$ acts centrally: $(c1_B)m = cm = m(c1_C)$ for $c \in \Bbbk$. From the perspective of general module theory over general rings (algebras over non-field commutative rings), we should have said that our bimodules are $\Bbbk$-central bimodule. Since we always work over a field, all bimodules we considered are always $\Bbbk$-central and we will not write such adjective everytime.*

Example 1.41.

- *Multiplication makes A_A a graded right module and ${}_AA_A$ a graded bimodule; each degree component is a bimodule ${}_{A_0}(A_i)_{A_0}$.*
- *For a bimodule ${}_AM_A$ and idempotents $e, f \in A$, the space $eMf := \{enf \mid n \in M\}$ is a bimodule ${}_{eAe}(eMf)_{fAf}$.*

Definition 1.42. *For a $\Bbbk$-algebra A, its enveloping algebra is $A^{\mathrm{en}} := A \otimes_{\Bbbk} A^{\mathrm{op}}$, with multiplication extended bilinearly from*

$$(a \otimes b^{\mathrm{op}})(a' \otimes (b')^{\mathrm{op}}) = aa' \otimes (b'b)^{\mathrm{op}}, \qquad 1_{A^{\mathrm{en}}} = 1_A \otimes 1_A^{\mathrm{op}} \quad (a, b, a', b' \in A).$$

A bimodule ${}_AM_A$ becomes a right A^{en}-module by

$$m \cdot (a \otimes b^{\mathrm{op}}) = bma \quad (a, b \in A,\ m \in M). \tag{1.8}$$

The first tensor factor records the right action, and the second the left action:

$$\big(m \cdot (a \otimes b^{\mathrm{op}})\big) \cdot (a' \otimes (b')^{\mathrm{op}}) = b'bmaa' = m \cdot \big(aa' \otimes (b'b)^{\mathrm{op}}\big).$$

Conversely, for $M \in \operatorname{Mod} A^{\mathrm{en}}$, define $ma = m \cdot (a \otimes 1)$ and $bm = m \cdot (1 \otimes b^{\mathrm{op}})$. The two actions commute because $a \otimes 1$ and $1 \otimes b^{\mathrm{op}}$ commute in A^{en}. For bimodules ${}_AM_A$, ${}_AN_A$, a linear map $u : M \to N$ satisfies

$$u(bma) = bu(m)a \quad (a, b \in A,\ m \in M) \quad \Longleftrightarrow \quad u \in \operatorname{Hom}_{A^{\mathrm{en}}}(M, N).$$

Thus $\operatorname{Mod} A^{\mathrm{en}}$ is also the category of A-bimodules and maps preserving both actions.

Grant–Iyama [GI20, §2, p. 2591] identify bimodules with left modules over A^{en}, using $(a \otimes b^{\mathrm{op}})m = amb$. Our right module convention is (1.8): $m \cdot (a \otimes b^{\mathrm{op}}) = bma$. Both describe the same bimodules; the side on which A^{en} acts determines the formula.

Remark 1.43. For a finite-dimensional semisimple $\Bbbk$-algebra S, semisimplicity of S^{en} requires an additional condition. For a field extension $K/\Bbbk$, give $K \otimes_{\Bbbk} S$ the multiplication $(c \otimes s)(d \otimes t) = cd \otimes st$. The algebra S is called separable over $\Bbbk$ if $K \otimes_{\Bbbk} S$ is semisimple for every field extension $K/\Bbbk$. By [DK94, Theorem 6.1.2, p. 105],

$$S^{\mathrm{en}} \text{ is semisimple} \quad \Longleftrightarrow \quad S \text{ is separable over } \Bbbk.$$

For a product of matrix algebras, every field extension $K/\Bbbk$ gives

$$K \otimes_{\Bbbk} \prod_{j=1}^{r} \operatorname{Mat}_{n_j}(\Bbbk) \xrightarrow{\sim} \prod_{j=1}^{r} \operatorname{Mat}_{n_j}(K), \qquad c \otimes (a_j)_j \longmapsto (ca_j)_j.$$

The target is semisimple by Theorem 1.12. Thus products of matrix algebras over $\Bbbk$ are separable; in particular, every finite-dimensional semisimple algebra over an algebraically closed field is separable. Semisimplicity of S alone does not imply semisimplicity of S^{en} over an arbitrary field.

Example 1.44. *Let $\Bbbk = \mathbb{F}_2(t)$, where t is an indeterminate, and let $S = \Bbbk[u]/(u^2 - t)$, also writing u for its residue class. The element t is not a square in $\Bbbk$: an equality $p(t)^2 = tq(t)^2$ for non-zero polynomials would give $2\deg p = 1 + 2\deg q$. Thus $u^2 - t$ is irreducible and S is a field, hence semisimple. However, since S is commutative and has characteristic 2,*

$$S^{\mathrm{en}} = S \otimes_{\Bbbk} S \cong S[v]/(v^2 - t) = S[v]/((v - u)^2) \cong S[\varepsilon]/(\varepsilon^2).$$

Here the first tensor factor supplies the coefficients, v corresponds to $1\otimes u$, and $\varepsilon = 1\otimes u - u\otimes 1$. The ideal $S\varepsilon$ is non-zero, has square zero, and is the unique maximal ideal, so $\operatorname{rad}(S^{\mathrm{en}}) = S\varepsilon \neq 0$. Thus S^{en} is not semisimple, and the field extension $S/\Bbbk$ already shows that S is not separable over $\Bbbk$.

1.10 Graded $\Bbbk$-linear duals

A linear functional on a graded vector space can be non-zero on infinitely many degrees. To obtain a graded space, retain only finite sums of homogeneous functionals.

Definition 1.45. *For a graded vector space $V = \bigoplus_{i\in\mathbb{Z}} V_i$, the graded $\Bbbk$-linear dual is the graded vector space*

$$\mathbb{D}(V) = \bigoplus_{i\in\mathbb{Z}} \mathbb{D}(V)_i, \qquad \mathbb{D}(V)_i = D(V_{-i}) = \operatorname{Hom}_{\Bbbk}^{\mathbb{Z}}(V, \Bbbk(i)).$$

Note that here $\Bbbk$ is regarded as a graded module concentrated in degree 0.

Linear duality reverses the side of a module action. For a right A-module M and a left A-module N, set

$$(af)(m) = f(ma), \qquad (ga)(n) = g(an) \quad \text{for } a\in A,\ f\in D(M),\ g\in D(N),\ m\in M,\ n\in N. \tag{1.9}$$

For graded modules these actions restrict to the graded linear duals: if $a\in A_p$ and $f\in\mathbb{D}(M)_q$, then af vanishes outside M_{-p-q}.

Example 1.46. *Let $A = \Bbbk A_3$ and $P_1 = e_{11}A$ as in Example 1.25. The functionals $\varepsilon_{1j}(e_{1\ell}) = \delta_{j\ell}$ for $1\le j,\ell\le 3$ form a basis of $\mathbb{D}(P_1)$, with*

$$\deg\varepsilon_{1j} = 1-j, \qquad e_{12}\varepsilon_{12} = \varepsilon_{11}, \qquad e_{23}\varepsilon_{13} = \varepsilon_{12}, \qquad e_{jj}\varepsilon_{1j} = \varepsilon_{1j}.$$

The actions follow from (1.9).

For any $u\in\operatorname{Hom}_A(M,N)$, we have a dual map

$$D(u) : D(N) \to D(M), \qquad f\mapsto f\circ u,$$

which is a left A-linear map, i.e. $D(u)\in\operatorname{Hom}_{A^{\mathrm{op}}}(D(N), D(M))$. This gives rise to four exact (contravariant) functors (without any finiteness assumption):

$$\begin{aligned} D &: (\operatorname{Mod} A)^{\mathrm{op}} \longrightarrow \operatorname{Mod}(A^{\mathrm{op}}), & D &: (\operatorname{Mod}(A^{\mathrm{op}}))^{\mathrm{op}} \longrightarrow \operatorname{Mod} A,\\ \mathbb{D} &: (\operatorname{GrMod} A)^{\mathrm{op}} \longrightarrow \operatorname{GrMod}(A^{\mathrm{op}}), & \mathbb{D} &: (\operatorname{GrMod}(A^{\mathrm{op}}))^{\mathrm{op}} \longrightarrow \operatorname{GrMod} A. \end{aligned} \tag{1.10}$$

Exactness follows from Example 1.10, degree by degree for $\mathbb{D}$.

For these to be duality, we need to recover a module from its bidual; this requires some finiteness conditions.

Lemma 1.47. *Let A be a $\Bbbk$-algebra.*

(1) *The functor D defines a duality between finite-dimensional right and left A-modules. In particular, if A is finite-dimensional, then we have a duality on finitely generated modules*

$$D : (\operatorname{mod} A)^{\mathrm{op}} \xrightarrow{\sim} \operatorname{mod}(A^{\mathrm{op}}), \qquad D : (\operatorname{mod}(A^{\mathrm{op}}))^{\mathrm{op}} \xrightarrow{\sim} \operatorname{mod} A.$$

(2) *Suppose that A is graded. For locally finite $M\in\operatorname{grmod} A$, evaluation is a natural graded module isomorphism*

$$\mathrm{ev}_M : M \xrightarrow{\sim} \mathbb{D}(\mathbb{D}(M)), \qquad m\longmapsto (f\mapsto f(m)). \tag{1.11}$$

In particular, we have a duality between locally finite graded right and left A-modules:

$$\mathbb{D} : (\operatorname{grmod} A)^{\mathrm{op}} \xrightarrow{\sim} \operatorname{grmod}(A^{\mathrm{op}}), \qquad \mathbb{D} : (\operatorname{grmod}(A^{\mathrm{op}}))^{\mathrm{op}} \xrightarrow{\sim} \operatorname{grmod} A. \tag{1.12}$$

Proof First we show that ev_M is A-linear. Indeed,

$$\mathrm{ev}_M(ma)(f) = f(ma) = \mathrm{ev}_M(m)(af) = (\mathrm{ev}_M(m)a)(f) \quad (m \in M,\ a \in A,\ f \in D(M)).$$

For (1), the claim follows from Example 1.10. For (2), apply (1) degree-wise. □

For a locally finite graded vector space V, restriction gives

$$\mathbb{D}(V) = \bigoplus_{i \in \mathbb{Z}} D(V_{-i}) \subset D(V) \cong \prod_{i \in \mathbb{Z}} D(V_i).$$

The inclusion is equality exactly when only finitely many V_i are non-zero. For an arbitrary vector space, evaluation $V \to D(D(V))$ is injective but need not be surjective; hence the functors in (1.10) are not dualities on all modules.

Example 1.48. *Graded linear duality need not preserve finite generation, even for a polynomial ring. Let $A = \Bbbk[x]$ with $\deg x = 1$ and let $\varepsilon_n(x^m) = \delta_{nm}$ for $n, m \geq 0$. Then $\deg \varepsilon_n = -n$, $x\varepsilon_0 = 0$, and $x\varepsilon_n = \varepsilon_{n-1}$ for $n \geq 1$. Any finite set of elements of $\mathbb{D}(A)$ lies in the span of $\varepsilon_0, \ldots, \varepsilon_N$ for some N, and multiplication by x preserves that span. Thus $\mathbb{D}(A)$ is locally finite but not finitely generated.*

1.11 Left, right, and two-sided duals over S

Fix a finite-dimensional semisimple $\Bbbk$-algebra S, which is regarded as concentrated in degree 0 when treated as a graded algebra. For a graded algebra A with $A_0 = S$, each degree component is a bimodule ${}_S(A_i)_S$.

Definition 1.49. *For a right S-module M and a left S-module N, define their S-duals by*

$$\begin{aligned} M^* &:= \operatorname{Hom}_S(M, S) \in \operatorname{Mod}(S^{\mathrm{op}}), \quad (sf)(m) := sf(m), \\ {}^*N &:= \operatorname{Hom}_{S^{\mathrm{op}}}(N, S) \in \operatorname{Mod} S, \qquad (gs)(n) := g(n)s. \end{aligned}$$

Here $s \in S$, $m \in M$, $n \in N$, $f \in M^*$, and $g \in {}^*N$.

Similar to $\Bbbk$-lienar duality, this definition extends to exact contravariant functors: for a right module map $u : M \to N$, define $u^* : N^* \to M^*$ by precomposition $u^*(f) = f \circ u$; for a left module map $u : M \to N$, the same recipe defines ${}^*u : {}^*N \to {}^*M$. This yields the desired functors

$$(-)^* : (\operatorname{Mod} S)^{\mathrm{op}} \longrightarrow \operatorname{Mod}(S^{\mathrm{op}}), \qquad {}^*(-) : (\operatorname{Mod}(S^{\mathrm{op}}))^{\mathrm{op}} \longrightarrow \operatorname{Mod} S.$$

We also have graded analogue similar to Definition 1.45.

Definition 1.50. *For graded right and left S-modules M, respectively, define their graded S-duals by*

$$M^{(*)} := \bigoplus_{i \in \mathbb{Z}} (M_{-i})^* = \bigoplus_{i \in \mathbb{Z}} \operatorname{Hom}_S^{\mathbb{Z}}(M, S(i)), \qquad {}^{(*)}M := \bigoplus_{i \in \mathbb{Z}} {}^*(M_{-i}) = \bigoplus_{i \in \mathbb{Z}} \operatorname{Hom}_{S^{\mathrm{op}}}^{\mathbb{Z}}(M, S(i)),$$

where each indicated summand has degree i.

This also extends to a functor: for a degree 0 right S-module map $u : M \to N$, define $u^{(*)} : N^{(*)} \to M^{(*)}$ by

$$(u^{(*)})_i : (N_{-i})^* \longrightarrow (M_{-i})^*, \qquad f \longmapsto f \circ u|_{M_{-i}} \quad (i \in \mathbb{Z}).$$

For a degree 0 left module map $u : M \to N$, the same formula defines ${}^{(*)}u : {}^{(*)}N \to {}^{(*)}M$. With the actions from Definition 1.49, we obtain functors

$$(-)^{(*)} : (\operatorname{GrMod} S)^{\mathrm{op}} \longrightarrow \operatorname{GrMod}(S^{\mathrm{op}}), \qquad {}^{(*)}(-) : (\operatorname{GrMod}(S^{\mathrm{op}}))^{\mathrm{op}} \longrightarrow \operatorname{GrMod} S. \tag{1.13}$$

Each functor in (1.13) restricts to locally finite modules: replace GrMod by grmod in its source and target. Indeed, for $R = S$ or S^{op} and $M \in \operatorname{grmod} R$,

$$\dim_{\Bbbk} \operatorname{Hom}_R(M_{-i}, R) \leq (\dim_{\Bbbk} M_{-i})(\dim_{\Bbbk} R) < \infty \quad (i \in \mathbb{Z}). \tag{1.14}$$

Lemma 1.51. *We have dualities*

$$(-)^* : (\operatorname{mod} S)^{\mathrm{op}} \xrightarrow{\sim} \operatorname{mod}(S^{\mathrm{op}}), \qquad {}^*(-) : (\operatorname{mod}(S^{\mathrm{op}}))^{\mathrm{op}} \xrightarrow{\sim} \operatorname{mod} S,$$
$$(-)^{(*)} : (\operatorname{grmod} S)^{\mathrm{op}} \xrightarrow{\sim} \operatorname{grmod}(S^{\mathrm{op}}), \quad {}^{(*)}(-) : (\operatorname{grmod}(S^{\mathrm{op}}))^{\mathrm{op}} \xrightarrow{\sim} \operatorname{grmod} S.$$

Proof We show that evaluation gives natural isomorphisms of functors

$$\mathrm{id}_{\operatorname{mod} S} \xrightarrow{\sim} {}^*((-)^*), \qquad \mathrm{id}_{\operatorname{mod}(S^{\mathrm{op}})} \xrightarrow{\sim} ({}^*(-))^*,$$

with components

$$\mathrm{ev}_M : M \xrightarrow{\sim} {}^*(M^*), \quad m \longmapsto (f \mapsto f(m)) \quad (M \in \operatorname{mod} S),$$
$$\mathrm{ev}_N : N \xrightarrow{\sim} ({}^*N)^*, \quad n \longmapsto (g \mapsto g(n)) \quad (N \in \operatorname{mod}(S^{\mathrm{op}})).$$

For $u : M \to N$ in $\operatorname{mod} S$, $m \in M$, and $f \in N^*$,

$$\big({}^*(u^*)(\mathrm{ev}_M(m))\big)(f) = (u^* f)(m) = f(u(m)) = \mathrm{ev}_N(u(m))(f).$$

Hence ${}^*(u^*) \circ \mathrm{ev}_M = \mathrm{ev}_N \circ u$. For $M = S$, evaluation is an isomorphism: its inverse sends $F \in {}^*(S^*)$ to $F(\mathrm{id}_S)$. Indeed, every $f \in S^*$ satisfies $f(s) = f(1)s$, so $F(f) = f(1)F(\mathrm{id}_S) = f(F(\mathrm{id}_S))$. Restriction to summands identifies the dual of a finite direct sum with the direct sum of the duals; evaluation therefore is an isomorphism for $S^{\oplus n}$. By Corollary 1.13, $M \oplus L \cong S^{\oplus n}$ for some L. Under the direct sum identifications, $\mathrm{ev}_{M\oplus L} = \mathrm{ev}_M \oplus \mathrm{ev}_L$, so ev_M is an isomorphism. Replacing S by S^{op} proves the left module statement.

For the graded cases, apply the ungraded case degree-wise. Note that since S is finite-dimensional, finitely generated S-module is the same as finite-dimensional S-modules. □

For a bimodule ${}_S M_S$, the linear and one-sided duals retain both actions:

$$(s\lambda t)(m) = \lambda(tms), \qquad (sft)(m) = sf(tm), \qquad (sgt)(m) = g(ms)t, \tag{1.15}$$

where $s, t \in S$, $m \in M$, $\lambda \in D(M)$, $f \in \operatorname{Hom}_S(M, S)$, and $g \in \operatorname{Hom}_{S^{\mathrm{op}}}(M, S)$. The functionals still take values in $\Bbbk$ or S. With precomposition on morphisms, we obtain

$$\begin{aligned} D,\ (-)^*,\ {}^*(-) &: (\operatorname{Mod} S^{\mathrm{en}})^{\mathrm{op}} \longrightarrow \operatorname{Mod} S^{\mathrm{en}}, \\ \mathbb{D},\ (-)^{(*)},\ {}^{(*)}(-) &: (\operatorname{GrMod} S^{\mathrm{en}})^{\mathrm{op}} \longrightarrow \operatorname{GrMod} S^{\mathrm{en}}. \end{aligned} \tag{1.16}$$

The restrictions to finite-dimensional modules, and to locally finite graded modules, are dualities by Lemmas 1.47 and 1.51.

Evaluation preserves both actions: for $m \in M$, $f \in M^*$, and $s, t \in S$,

$$(s\,\mathrm{ev}_M(m)\,t)(f) = f(sm)t = f(smt) = \mathrm{ev}_M(smt)(f).$$

The corresponding checks for D and the left dual use the same evaluation formula; applying them in each degree gives the graded statements.

Recall from Definition 1.42 and (1.8) that $S^{\mathrm{en}} = S \otimes_{\Bbbk} S^{\mathrm{op}}$ and S-bimodules are the same thing as S^{en}-modules via $m \cdot (s \otimes t^{\mathrm{op}}) = tms$.

Definition 1.52. *The two-sided dual of a bimodule ${}_SM_S$ is*

$$M^{*\mathrm{en}} := \mathrm{Hom}_{S^{\mathrm{en}}}(M, S^{\mathrm{en}}), \qquad (sFt)(m) := (s \otimes t^{\mathrm{op}})F(m) \quad (s, t \in S,\ F \in M^{*\mathrm{en}},\ m \in M).$$

In Definition 1.52, M has the right action (1.8), so $F(tms) = F(m)(s \otimes t^{\mathrm{op}})$. The output is a left S^{en}-module by multiplication on the values, which gives the displayed bimodule action. The two-sided dual takes values in S^{en}, not in S. On bimodule morphisms it acts by precomposition, as do D, $(-)^*$, and ${}^*(-)$; each is a contravariant functor on the category of S-bimodules. The two-sided dual records both module actions in the target algebra S^{en}.

For a graded bimodule ${}_SM_S$, define its *graded two-sided dual* by

$$M^{(*\mathrm{en})} := \bigoplus_{i \in \mathbb{Z}} (M_{-i})^{*\mathrm{en}}, \qquad (M^{(*\mathrm{en})})_i = (M_{-i})^{*\mathrm{en}}.$$

Definition 1.52 gives a functor

$$(-)^{(*\mathrm{en})} : (\mathrm{GrMod}\, S^{\mathrm{en}})^{\mathrm{op}} \longrightarrow \mathrm{GrMod}\, S^{\mathrm{en}}. \tag{1.17}$$

On a degree 0 bimodule map $u : M \to N$, its degree i component is

$$(u^{(*\mathrm{en})})_i : (N_{-i})^{*\mathrm{en}} \longrightarrow (M_{-i})^{*\mathrm{en}}, \qquad F \longmapsto F \circ u|_{M_{-i}} \quad (i \in \mathbb{Z}).$$

The functor restricts to locally finite bimodules by (1.14) with $R = S^{\mathrm{en}}$.

The four duals above follow the distinction in Grant–Iyama [GI20, §2.1, pp. 2592–2593], with the following notation changes:

construction	Grant–Iyama	these notes
$\Bbbk$-linear dual	M^*	$D(M)$
left S-linear dual	$M^{*\ell}$	${}^*M$
right S-linear dual	M^{*r}	M^*
two-sided dual	$M^\vee$	$M^{*\mathrm{en}}$

The last row also changes the convention for enveloping modules. Their $F : M \to S^{\mathrm{en}}$ is left S^{en}-linear. Define

$$\sigma : S^{\mathrm{en}} \longrightarrow S^{\mathrm{en}}, \qquad a \otimes b^{\mathrm{op}} \longmapsto b \otimes a^{\mathrm{op}}.$$

Since $\sigma(xy) = \sigma(y)\sigma(x)$ for $x, y \in S^{\mathrm{en}}$, the map $\sigma \circ F : M \to S^{\mathrm{en}}$ satisfies

$$(\sigma \circ F)(tms) = (\sigma \circ F)(m)(s \otimes t^{\mathrm{op}}) \quad (s, t \in S,\ m \in M),$$

and belongs to our $M^{*\mathrm{en}}$. Thus $M^\vee \to M^{*\mathrm{en}}$, $F \mapsto \sigma \circ F$, translates their two-sided dual into our convention. In particular, their star denotes the linear dual; our star denotes the right S-dual.

1.12 Comparing the duals

The one-sided S-duals and the $\Bbbk$-linear dual are naturally isomorphic after choosing a symmetrising form on S.

Definition 1.53. *A symmetrising form on a finite-dimensional $\Bbbk$-algebra S is a linear map $\tau : S \to \Bbbk$ satisfying*

$$\tau(st) = \tau(ts) \quad \text{for all } s,t \in S, \qquad \big(\tau(st) = 0 \text{ for all } t \in S\big) \implies s = 0.$$

An algebra admitting such a form is symmetric.

The second condition in Definition 1.53 says that the bilinear form $(s,t) \mapsto \tau(st)$ is non-degenerate: every non-zero $s \in S$ pairs non-trivially with some $t \in S$.

Every finite-dimensional semisimple $\Bbbk$-algebra admits a symmetrising form. For products of matrix algebras over $\Bbbk$, the forms are explicit:

Example 1.54. *The following maps are symmetrising forms:*

$$\tau : \Bbbk^{\oplus r} \longrightarrow \Bbbk, \quad (a_1, \dots, a_r) \longmapsto \sum_{j=1}^{r} a_j,$$

$$\tau : \mathrm{Mat}_n(\Bbbk) \longrightarrow \Bbbk, \qquad a \longmapsto \mathrm{Tr}(a) := \sum_{i=1}^{n} a_{ii},$$

and more generally, $$\tau : \prod_{j=1}^{r} \mathrm{Mat}_{n_j}(\Bbbk) \longrightarrow \Bbbk, \quad (a_1, \dots, a_r) \longmapsto \sum_{j=1}^{r} \mathrm{Tr}(a_j).$$

Generally, any finite-dimensional semisimple algebra has a symmetrising form, by Eilenberg–Nakayama.[a] For example, if $\lambda_j : \Delta_j \to \Bbbk$ is a symmetrising form for a finite-dimensional division $\Bbbk$-algebra Δ_j with $j = 1, \dots, n$. Then the semisimple algebra $S = \prod_{j=1}^{r} \mathrm{Mat}_{n_j}(\Delta_j)$ has a symmetrising form given by

$$\tau : S \longrightarrow \Bbbk, \qquad (a_1, \dots, a_r) \longmapsto \sum_{j=1}^{r} \sum_{i=1}^{n_j} \lambda_j((a_j)_{ii}).$$

[a] S. Eilenberg and T. Nakayama, *On the dimension of modules and algebras, II. Frobenius algebras and quasi-Frobenius rings*, Nagoya Math. J. **9** (1955), 1–16, §2, Propositions 3–5, pp. 4–5; doi:10.1017/S0027763000023229.

Theorem 1.55 (Comparison of duals). *Choose a symmetrising form $\tau : S \to \Bbbk$. Postcomposition with τ defines natural isomorphisms from the S-dual (on either side) to D, and also from the graded S-dual (on either side) to $\mathbb{D}$.*

We can spell this out more precisely:

Ungraded duals. For $M \in \operatorname{mod} S$ and $N \in \operatorname{mod}(S^{\mathrm{op}})$, the maps

$$\alpha_M : M^* \longrightarrow D(M), \quad f \longmapsto \tau \circ f, \qquad \beta_N : {}^*N \longrightarrow D(N), \quad g \longmapsto \tau \circ g \tag{1.18}$$

are S-module isomorphisms and give natural isomorphisms

$$\begin{aligned} \alpha &: (-)^* \xrightarrow{\sim} D \quad \text{of functors } (\operatorname{mod} S)^{\mathrm{op}} \longrightarrow \operatorname{mod}(S^{\mathrm{op}}), \\ \beta &: {}^*(-) \xrightarrow{\sim} D \quad \text{of functors } (\operatorname{mod}(S^{\mathrm{op}}))^{\mathrm{op}} \longrightarrow \operatorname{mod} S. \end{aligned} \tag{1.19}$$

Graded duals. Composition with τ in each degree gives natural isomorphisms

$$\begin{aligned} \alpha &: (-)^{(*)} \xrightarrow{\sim} \mathbb{D} \quad \text{of functors } (\operatorname{grmod} S)^{\mathrm{op}} \longrightarrow \operatorname{grmod}(S^{\mathrm{op}}), \\ \beta &: {}^{(*)}(-) \xrightarrow{\sim} \mathbb{D} \quad \text{of functors } (\operatorname{grmod}(S^{\mathrm{op}}))^{\mathrm{op}} \longrightarrow \operatorname{grmod} S. \end{aligned} \tag{1.20}$$

For $M \in \operatorname{grmod} S$ and $N \in \operatorname{grmod}(S^{\mathrm{op}})$, the degree i components are $\alpha_{M_{-i}}$ and $\beta_{N_{-i}}$, respectively.

Proof We prove the right module statement, i.e. taking $\alpha_M(f) := \tau \circ f$ yields

$$\text{ungraded case } \alpha : (-)^* \xrightarrow{\sim} D, \quad \text{and} \quad \text{graded case } \alpha : (-)^{(*)} \xrightarrow{\sim} \mathbb{D}.$$

First, non-degeneracy of τ gives an isomorphism $S \to D(S)$, $a \mapsto (s \mapsto \tau(as))$. Hence, for $\lambda \in D(M)$, there is a unique linear map $f : M \to S$ satisfying

$$\tau(f(m)s) = \lambda(ms) \quad \text{for all } m \in M,\ s \in S. \tag{1.21}$$

For $m \in M$ and $s, t \in S$,

$$\tau(f(mt)s) = \lambda(mts) = \tau(f(m)ts),$$

so non-degeneracy implies $f(mt) = f(m)t$. Taking $s = 1$ in (1.21) gives $\tau \circ f = \lambda$; conversely, every right S-linear preimage of λ must satisfy (1.21). Thus α_M is bijective.

Both M^* and $D(M)$ are left S-modules, so the component $\alpha_M : M^* \to D(M)$ must preserve the left action:

$$\alpha_M(sf)(m) = \tau(sf(m)) = \tau(f(m)s) = \tau(f(ms)) = (s\alpha_M(f))(m) \quad (s \in S,\ f \in M^*,\ m \in M).$$

For a right module map $u : M \to N$ and $f \in N^*$, associativity of composition gives $\alpha_M(f \circ u) = \alpha_N(f) \circ u$, proving naturality.

For the graded comparisons, apply the ungraded natural isomorphism degree-wise.

Same argument applies dually to S^{op} and β. □

Remark 1.56. The natural isomorphisms in Theorem 1.55 depend on the chosen symmetrising form. For example, every choice $c_1, \dots, c_r \in \Bbbk \setminus \{0\}$ gives a form $\tau(a_1, \dots, a_r) = \sum_{i=1}^r c_i a_i$ on $\Bbbk^{\oplus r}$.

Remark 1.57. Let A be a graded algebra with $A_0 = S$. The space $\operatorname{Hom}_S(M, S)$ do not have a natural left A-module on the spot, but can be after *picking a symmetrising form τ of S*. Indeed, the comparison (1.20) transports the canonical left A-action on $\mathbb{D}(M)$ to the graded S-dual of a locally finite module:

$$af = \alpha_M^{-1}(a\alpha_M(f)) \quad \text{for right } M, \qquad ga = \beta_N^{-1}(\beta_N(g)a) \quad \text{for left } N,$$

where $a \in A$, $f \in M^{(*)}$, and $g \in {}^{(*)}N$. Similarly for right action on left dual.

The comparison theorem has a bimodule version:

Theorem 1.58. *With the actions in* (1.15)*, we obtain natural isomorphisms of functors*

$$\begin{aligned} (-)^* \underset{\alpha}{\xrightarrow{\sim}} D \underset{\beta}{\xleftarrow{\sim}} {}^*(-) \quad &: (\operatorname{mod} S^{\mathrm{en}})^{\mathrm{op}} \longrightarrow \operatorname{mod} S^{\mathrm{en}}, \\ (-)^{(*)} \underset{\alpha}{\xrightarrow{\sim}} \mathbb{D} \underset{\beta}{\xleftarrow{\sim}} {}^{(*)}(-) \quad &: (\operatorname{grmod} S^{\mathrm{en}})^{\mathrm{op}} \longrightarrow \operatorname{grmod} S^{\mathrm{en}}. \end{aligned} \tag{1.22}$$

There is also a natural isomorphism $\gamma : (-)^{*_{\mathrm{en}}} \xrightarrow{\sim} (-)^*$ given by

$$\gamma_M : M^{*_{\mathrm{en}}} \to M^*, \qquad F \mapsto \big(m \mapsto \sum_j a_j \tau(b_j)\big) \quad \text{for} \quad F(m) = \sum_j a_j \otimes b_j^{\mathrm{op}}, \tag{1.23}$$

which also depends on τ, but does not require S^{en} to be semisimple.

Consequently, on finite-dimensional S-bimodules all four dual functors are naturally isomorphic:

$$(-)^{*_{\mathrm{en}}} \underset{\gamma}{\xrightarrow{\sim}} (-)^* \underset{\alpha}{\xrightarrow{\sim}} D \underset{\beta}{\xleftarrow{\sim}} {}^*(-).$$

Applying γ in each degree gives the corresponding natural isomorphisms on locally finite graded S-bimodules:

$$(-)^{(*_{\mathrm{en}})} \underset{\gamma}{\xrightarrow{\sim}} (-)^{(*)} \underset{\alpha}{\xrightarrow{\sim}} \mathbb{D} \underset{\beta}{\xleftarrow{\sim}} {}^{(*)}(-).$$

The first chain consists of functors $(\operatorname{mod} S^{\mathrm{en}})^{\mathrm{op}} \to \operatorname{mod} S^{\mathrm{en}}$; the second consists of functors $(\operatorname{grmod} S^{\mathrm{en}})^{\mathrm{op}} \to \operatorname{grmod} S^{\mathrm{en}}$.

Proof The form $\tau_{\mathrm{en}}(a \otimes b^{\mathrm{op}}) = \tau(a)\tau(b)$ is symmetrising on S^{en}. Indeed, $\tau_{\mathrm{en}}((a \otimes b^{\mathrm{op}})(c \otimes d^{\mathrm{op}})) = \tau(ac)\tau(db) = \tau(ca)\tau(bd)$, and its pairing is the tensor product of two non-degenerate pairings. The formula (1.23) satisfies

$$\gamma_M(F)(ms) = \gamma_M(F)(m)s, \qquad \gamma_M(sFt) = s\gamma_M(F)t, \qquad \alpha_M(\gamma_M(F)) = \tau_{\mathrm{en}} \circ F.$$

The argument (1.21), applied to S^{en} and τ_{en}, makes $F \mapsto \tau_{\mathrm{en}} \circ F$ an isomorphism. Thus γ_M is an isomorphism. For a bimodule map $u : M \to N$ and $F \in N^{*_{\mathrm{en}}}$, the formula (1.23) gives $\gamma_M(F \circ u) = \gamma_N(F) \circ u$, proving naturality. Taking the direct sum of $\gamma_{M_{-i}}$ over $i \in \mathbb{Z}$ gives the graded two-sided comparison. Each component is a bimodule isomorphism, and naturality follows in each degree from the ungraded comparison. □

1.13 Exercises

Exercise 1.59 (Matrix calculations). *Verify the grading on $\Bbbk A_n$ in Example 1.19 using $e_{ij}e_{pq} = \delta_{jp}e_{iq}$. Compute $\operatorname{rad}(\Bbbk A_3)$ from the definition.* Hint. *The strictly upper triangular ideal J satisfies $J^3 = 0$. Apply the argument of Example 1.14 to simple modules and to $A/J \cong \Bbbk^{\oplus 3}$.*

Exercise 1.60 (Shifts and duals). *For $A = \Bbbk A_3$, compute the degrees and all non-zero matrix-unit actions on $\mathbb{D}(e_{22}A)$. Show that it is not isomorphic to any left module Ae_{jj}, even after a shift.* Hint. *Compare dimensions and the actions of e_{11}, e_{22}, e_{33}.*

Exercise 1.61 (Homogeneous maps). *The finite-generation hypothesis in Lemma 1.31 is needed. Let $S = \Bbbk e_1 \oplus \Bbbk e_2$ with componentwise multiplication, let $N = e_1S$ have degree 0, and put*

$$M = \bigoplus_{i \geq 0} (e_1S)(-i) = \bigoplus_{i \geq 0} \Bbbk m_i, \qquad \deg m_i = i, \quad m_ie_1 = m_i, \quad m_ie_2 = 0.$$

For the S-linear map $f : M \to N$ with $f(m_i) = e_1$ for every $i \geq 0$, determine all homogeneous components and prove that f is not a finite sum of homogeneous maps.

Exercise 1.62 (Dependence on the form). *Let $S = (\Bbbk A_2)_0 = \Bbbk e_{11} \oplus \Bbbk e_{22}$, $V = (\Bbbk A_2)_1 = \Bbbk e_{12}$, and $c_1, c_2 \in \Bbbk \setminus \{0\}$. Set $\tau(ae_{11} + be_{22}) = c_1a + c_2b$ and let $\varepsilon : V \to \Bbbk$ satisfy $\varepsilon(e_{12}) = 1$. Find the right and left S-linear maps corresponding to ε under* (1.18)*, and determine how they depend on c_1, c_2.*

Exercise 1.63 (Dependence of the action on the form). *Assume* $\operatorname{char}\Bbbk \neq 2$. *This exercise compares the A-actions transported through two symmetrising forms as in Remark 1.57. Let $A = \Bbbk A_2$, $S = \Bbbk e_{11} \oplus \Bbbk e_{22}$, and $M = e_{11}A$. For $c_1, c_2 \in \Bbbk \setminus \{0\}$, define*

$$\tau : S \longrightarrow \Bbbk, \qquad ae_{11} + be_{22} \longmapsto c_1 a + c_2 b \quad (a, b \in \Bbbk). \tag{1.24}$$

Consider the right S-linear maps

$$\begin{aligned} &\varphi_0 : M \longrightarrow S, \quad ae_{11} + be_{12} \longmapsto ae_{11}, \\ &\varphi_1 : M \longrightarrow S, \quad ae_{11} + be_{12} \longmapsto be_{22} \quad (a, b \in \Bbbk). \end{aligned} \tag{1.25}$$

(1) *Verify that τ is symmetrising and that φ_0, φ_1 form a homogeneous basis of $M^{(*)}$. Determine their degrees.*

(2) *Set $\tau_1(ae_{11} + be_{22}) = a + b$ and $\tau_2(ae_{11} + be_{22}) = a + 2b$. Write $\cdot_1, \cdot_2$ for the respective left A-actions transported from $\mathbb{D}(M)$ using* (1.18). *Calculate the actions of e_{11}, e_{22}, e_{12} on φ_0, φ_1 for both forms. Are the two actions equal?*

(3) *Construct a graded A-module isomorphism*

$$t : (M^{(*)}, \cdot_1) \longrightarrow (M^{(*)}, \cdot_2)$$

by giving $t(\varphi_0)$ and $t(\varphi_1)$.

2 Tensor and quadratic algebras

2.1 Tensor products over an algebra

Let A be any $\Bbbk$-algebra, M a right A-module, and N a left A-module. The tensor product over A imposes the identities $ma \otimes n = m \otimes an$ for $m \in M$, $n \in N$, and $a \in A$.

Definition 2.1. *The tensor product of M and N over A, denoted $M \otimes_A N$, is the quotient of the vector space $M \otimes_\Bbbk N$ by*

$$\operatorname{span}_\Bbbk\{ma \otimes n - m \otimes an \mid m \in M,\ n \in N,\ a \in A\}.$$

By abuse of notation, we write $m \otimes n$ for the its own image in the $M \otimes_\Bbbk N \to M \otimes_A N$; if clarity is needed, then we will write $m \otimes_A n$.

Lemma 2.2 (Basic properties of tensor products). *Let A be any $\Bbbk$-algebra, M a right A-module, and N a left A-module.*

(1) *For a vector space W and a bilinear map $b : M \times N \to W$ satisfying $b(ma, n) = b(m, an)$ for all $m \in M$, $a \in A$, and $n \in N$, there is a unique linear factorisation (commutative diagram)*

$$\begin{array}{ccc} M \times N & \xrightarrow{(m,n)\mapsto m\otimes_A n} & M \otimes_A N \\ & \searrow^{\forall\ \text{bilinear } b} & \downarrow \exists!\, \widetilde{b} \\ & & W. \end{array}$$

(2) *If B, C are $\Bbbk$-algebras and ${}_BM_A$, ${}_AN_C$ are bimodules, then $M \otimes_A N$ is a bimodule ${}_B(M \otimes_A N)_C$ with*

$$b(m \otimes_A n)c = bm \otimes_A nc \qquad (b \in B,\ c \in C,\ m \in M,\ n \in N).$$

For right A-module maps $u : M \to M'$ and left A-module maps $v : N \to N'$, the assignment $(m, n) \mapsto u(m) \otimes v(n)$ induces

$$u \otimes_A v : M \otimes_A N \longrightarrow M' \otimes_A N', \qquad m \otimes n \longmapsto u(m) \otimes v(n).$$

These define functors

$$(-) \otimes_A N : \operatorname{Mod} A \longrightarrow \operatorname{Mod} C, \qquad M \otimes_A (-) : \operatorname{Mod}(A^{\mathrm{op}}) \longrightarrow \operatorname{Mod}(B^{\mathrm{op}}).$$

On bimodule maps, they retain the outer actions:

$$(-) \otimes_A N : \operatorname{Mod} A \otimes_\Bbbk B^{\mathrm{op}} \longrightarrow \operatorname{Mod} C \otimes_\Bbbk B^{\mathrm{op}}, \qquad M \otimes_A (-) : \operatorname{Mod} C \otimes_\Bbbk A \longrightarrow \operatorname{Mod} C \otimes_\Bbbk B.$$

(3) *For $\Bbbk$-algebras A, B, a right A-module U, a bimodule ${}_AV_B$, and a left B-module W, there is a natural linear isomorphism*

$$(U \otimes_A V) \otimes_B W \xrightarrow{\sim} U \otimes_A (V \otimes_B W), \qquad (u \otimes v) \otimes w \longmapsto u \otimes (v \otimes w).$$

There are also natural module isomorphisms

$$M \otimes_A A \xrightarrow{\sim} M, \quad m \otimes a \longmapsto ma, \qquad A \otimes_A N \xrightarrow{\sim} N, \quad a \otimes n \longmapsto an.$$

Moreover, these maps are linear with respect to any additional bimodule action on U, W, M, N.

(4) *Tensor products distribute over direct sums in either variable. For arbitrary families $(M_i)_{i\in I}$ of right A-modules and $(N_j)_{j\in J}$ of left A-modules, indexed by sets I, J,*

$$\Big(\bigoplus_{i\in I} M_i\Big) \otimes_A N \xrightarrow{\sim} \bigoplus_{i\in I} (M_i \otimes_A N), \quad (m_i)_{i\in I} \otimes n \longmapsto (m_i \otimes n)_{i\in I},$$
$$M \otimes_A \Big(\bigoplus_{j\in J} N_j\Big) \xrightarrow{\sim} \bigoplus_{j\in J} (M \otimes_A N_j), \quad m \otimes (n_j)_{j\in J} \longmapsto (m \otimes n_j)_{j\in J}.$$

Proof For (1), Proposition 1.37 gives a unique linear map $M \otimes_{\Bbbk} N \to W$, $m \otimes n \mapsto b(m,n)$. The map vanishes on the defining subspace precisely when $b(ma,n) = b(m,an)$.

For (2), the outer actions preserve that subspace: for $b \in B$, $c \in C$, and $a \in A$,

$$b(ma \otimes n - m \otimes an)c = (bm)a \otimes nc - bm \otimes a(nc).$$

The module axioms follow from those on M and N. For module maps u, v,

$$u(ma) \otimes v(n) = u(m)a \otimes v(n) = u(m) \otimes av(n) = u(m) \otimes v(an),$$

so the displayed formula induces $u \otimes_A v$. The functor identities follow from $(u' \otimes v')(u(m) \otimes v(n)) = u'u(m) \otimes v'v(n)$ and $(\mathrm{id}_M \otimes \mathrm{id}_N)(m \otimes n) = m \otimes n$. If u, v preserve the outer actions, then $u \otimes_A v$ does as well.

For (3), both parenthesisations impose

$$(ua) \otimes v \otimes w = u \otimes (av) \otimes w, \qquad u \otimes (vb) \otimes w = u \otimes v \otimes (bw) \quad (a \in A,\ b \in B).$$

Thus the displayed associativity map and the reverse assignment are well-defined and inverse. The unit inverses are $M \to M \otimes_A A$, $m \mapsto m \otimes 1$, and $N \to A \otimes_A N$, $n \mapsto 1 \otimes n$.

For (4), the bilinear map $((m_i)_{i\in I}, n) \mapsto (m_i \otimes n)_{i\in I}$ satisfies the identity in (1), so it induces the displayed linear map. For each $i \in I$, let $\iota_i : M_i \to \bigoplus_{j\in I} M_j$ be the inclusion. The inverse sends $m \otimes n$ in the ith summand to $\iota_i(m) \otimes n$ and extends over the direct sum. Both composites fix every element $m \otimes n$ from a summand. The other variable is handled in the same way. □

For finite sets I, J and right A-modules M_j, N_i, there is a natural isomorphism

$$\begin{aligned} \mathrm{Hom}_A\Big(\bigoplus_{j\in J} M_j, \bigoplus_{i\in I} N_i\Big) &\xrightarrow{\sim} \bigoplus_{\substack{i\in I\\ j\in J}} \mathrm{Hom}_A(M_j, N_i), \\ f &\longmapsto (\pi_i f \iota_j)_{i\in I,\ j\in J}, \end{aligned} \tag{2.1}$$

where $\iota_j : M_j \to \bigoplus_{j'\in J} M_{j'}$ and $\pi_i : \bigoplus_{i'\in I} N_{i'} \to N_i$ are the inclusions and projections.

Lemma 2.3 (Hom, direct sums, and direct products). *Let M, N be right A-modules, and let $(M_i)_{i\in I}$ and $(N_i)_{i\in I}$ be arbitrary families of right A-modules, indexed by a set I. Write $\iota_i : M_i \to \bigoplus_{j\in I} M_j$ for the inclusions and $\pi_i : \prod_{j\in I} N_j \to N_i$ for the projections. There are natural linear isomorphisms*

$$\mathrm{Hom}_A\Big(\bigoplus_{i\in I} M_i, N\Big) \xrightarrow{\sim} \prod_{i\in I} \mathrm{Hom}_A(M_i, N), \quad f \longmapsto (f \circ \iota_i)_{i\in I},$$
$$\mathrm{Hom}_A\Big(M, \prod_{i\in I} N_i\Big) \xrightarrow{\sim} \prod_{i\in I} \mathrm{Hom}_A(M, N_i), \quad f \longmapsto (\pi_i \circ f)_{i\in I}.$$

There is also a natural injection

$$\bigoplus_{i\in I} \mathrm{Hom}_A(M, N_i) \longrightarrow \mathrm{Hom}_A\Big(M, \bigoplus_{i\in I} N_i\Big), \qquad (f_i)_{i\in I} \longmapsto \big(m \mapsto (f_i(m))_{i\in I}\big),$$

which is an isomorphism when M is finitely generated.

Proof The inverse of the first map sends $(f_i)_{i\in I}$ to $(m_i)_{i\in I} \mapsto \sum_{i\in I} f_i(m_i)$; the sum is finite because $(m_i)_{i\in I}$ has finite support. The inverse of the second sends $(f_i)_{i\in I}$ to $m \mapsto (f_i(m))_{i\in I}$; a direct product imposes no support condition. For the third map, coordinate projections recover each f_i, proving injectivity. If $m_1, \dots, m_r$ generate M, the images of these generators under $f : M \to \bigoplus_{i\in I} N_i$ are supported on a common finite subset $J \subset I$. Then $f(M) \subset \bigoplus_{i\in J} N_i$, so f comes from a finite family of maps. All three constructions commute with precomposition and postcomposition. □

Proposition 2.4 (Tensor products and direct products). *Let A be a $\Bbbk$-algebra, M, M_i right A-modules, and N, N_i left A-modules, with $i \in I$. The coordinate projections define linear maps*

$$\begin{aligned} \kappa_N &: \Big(\prod_i M_i\Big) \otimes_A N \longrightarrow \prod_i (M_i \otimes_A N), \quad (m_i)_i \otimes n \longmapsto (m_i \otimes n)_i, \\ \lambda_M &: M \otimes_A \Big(\prod_i N_i\Big) \longrightarrow \prod_i (M \otimes_A N_i), \quad m \otimes (n_i)_i \longmapsto (m \otimes n_i)_i. \end{aligned} \tag{2.2}$$

(1) *If I is finite, both maps are isomorphisms.*

(2) *The map κ_N is an isomorphism if N is a direct summand of $A^{\oplus r}$ as a left module for some finite r. The corresponding condition on the right module M makes λ_M an isomorphism.*

(3) *If A is finite-dimensional semisimple, both maps are injective. Moreover, κ_N is an isomorphism when N is finite-dimensional, and λ_M is an isomorphism when M is finite-dimensional.*

Proof The formulas in (2.2) are bilinear. For $s \in A$, the component identities $(m_i s) \otimes n = m_i \otimes sn$ and $(ms) \otimes n_i = m \otimes sn_i$ show that both formulas give well-defined maps from the indicated tensor products. For finite I, products equal sums, and Lemma 2.2(4) proves (1).

For $N = A$, the map κ_A is the identity on $\prod_i M_i$ under $m \otimes s \mapsto ms$. For $N = A^{\oplus r}$, both sides are $(\prod_i M_i)^{\oplus r}$, since a finite direct sum commutes with a direct product. If $N \oplus N' \cong A^{\oplus r}$, the comparison for $N \oplus N'$ is $\kappa_N \oplus \kappa_{N'}$. Its bijectivity implies the bijectivity of κ_N. This proves the assertion about κ_N in (2).

Now let A be finite-dimensional semisimple. For finite-dimensional N, Corollary 1.13 gives (2). For arbitrary N, take $z = \sum_{\ell=1}^{r} m^{(\ell)} \otimes n_\ell$ in $\operatorname{Ker} \kappa_N$ and set $N_0 = \sum_{\ell=1}^{r} An_\ell$. The submodule N_0 is finite-dimensional and has a complement in N by Theorem 1.12. Thus the inclusions

$$\Big(\prod_i M_i\Big) \otimes_A N_0 \longrightarrow \Big(\prod_i M_i\Big) \otimes_A N, \qquad M_i \otimes_A N_0 \longrightarrow M_i \otimes_A N$$

are injective, by the direct sum formula in Lemma 2.2(4). Regard z as an element with second factors in N_0. The equality $\kappa_N(z) = 0$ then gives $\kappa_{N_0}(z) = 0$. Aince κ_{N_0} is an isomorphism, $z = 0$.

For λ_M, apply the same arguments over A^{op}, using $N \otimes_{A^{\mathrm{op}}} M \cong M \otimes_A N$, $n \otimes m \mapsto m \otimes n$. □

The finite-dimensional hypothesis on the fixed factor cannot be dropped from the isomorphism assertion: Example 1.39 gives a non-surjective comparison even for vector spaces. Thus tensor products always distribute over direct sums, but not over arbitrary direct products.

Remark 2.5. For infinite families, Hom need not preserve direct sums in its second variable. For $M = \bigoplus_{i\geq 1} \Bbbk v_i$, the identity $M \to M$ does not come from $\bigoplus_i \operatorname{Hom}_\Bbbk(M, \Bbbk v_i)$: its projection onto every $\Bbbk v_i$ is non-zero. Thus the injection in Lemma 2.3 need not be surjective.

A product in the source of Hom behaves differently from a product in the target. For example, a linear map $\prod_{i\geq 1}\Bbbk \to \Bbbk$ need not be determined by its restrictions to the coordinate copies of $\Bbbk$. Indeed, $(1,1,\ldots) \notin \bigoplus_{i\geq 1}\Bbbk$. Define a functional on $\bigoplus_{i\geq 1}\Bbbk + \Bbbk(1,1,\ldots)$ to vanish on the direct sum and take value 1 at $(1,1,\ldots)$, and extend it to the product by extending a basis. The resulting non-zero functional vanishes on every coordinate copy.

For a bimodule ${}_AV_A$, define

$$V^{\otimes_A 0} := A, \qquad V^{\otimes_A i} := \underbrace{V \otimes_A \cdots \otimes_A V}_{i \text{ factors}} \quad (i \geq 1).$$

Note that there is no parenthesis in the definition of $V^{\otimes_A i}$ since Lemma 2.2(3) says that tenso product is an associatitive operation.

Let A, B, C be $\Bbbk$-algebras. For bimodules ${}_AM_B$, ${}_CX_B$, we have $\operatorname{Hom}_B(M,X) \in C\operatorname{Mod} A$, with

$$(cfa)(m) = cf(am) \quad (f \in \operatorname{Hom}_B(M,X),\ c \in C,\ a \in A,\ m \in M). \tag{2.3}$$

If $X \in \operatorname{Mod} B$ has no specified left action, the same formula defines the right A-action $(fa)(m) = f(am)$. The left-module version follows by passing to opposite algebras.

Proposition 2.6 (Tensor and Hom). *Consider bimodules ${}_AM_B$, ${}_BN_C$, a right C-module X, and a left A-module Y. There are maps, natural in M, N,*

$$\begin{aligned} \theta_r : \operatorname{Hom}_C(N,X) \otimes_B \operatorname{Hom}_B(M,B) &\longrightarrow \operatorname{Hom}_C(M \otimes_B N, X), \\ \theta_r(g \otimes f)(m \otimes n) &= g(f(m)n), \\ \theta_\ell : \operatorname{Hom}_{B^{\mathrm{op}}}(N,B) \otimes_B \operatorname{Hom}_{A^{\mathrm{op}}}(M,Y) &\longrightarrow \operatorname{Hom}_{A^{\mathrm{op}}}(M \otimes_B N, Y), \\ \theta_\ell(g \otimes f)(m \otimes n) &= f(mg(n)). \end{aligned} \tag{2.4}$$

The maps θ_r and θ_ℓ are right A-linear and left C-linear, respectively.

(1) *The map θ_r is an isomorphism if $M \in \operatorname{add} B$ as a right B-module.*

(2) *The map θ_ℓ is an isomorphism if ${}_BN \in \operatorname{add}_{\operatorname{mod} B^{\mathrm{op}}}(B)$.*

Here $\operatorname{add} B = \operatorname{add}_{\operatorname{mod} B}(B)$, as in Example 1.5. If ${}_CX_C$ and ${}_AY_A$ are bimodules, the respective maps also preserve the additional actions and are morphisms in $C\operatorname{Mod} A$.

Proof *Well-definedness of θ_r.* For $f \in \operatorname{Hom}_B(M,B)$, $g \in \operatorname{Hom}_C(N,X)$, and $b \in B$,

$$g(f(mb)n) = g(f(m)bn), \qquad (gb)(f(m)n) = g(bf(m)n) = g((bf)(m)n).$$

These identities respect the defining identifications in both tensor products. Right C-linearity follows from $g(f(m)nc) = g(f(m)n)c$ for $c \in C$. For $a \in A$,

$$\theta_r(g \otimes fa)(m \otimes n) = g(f(am)n) = (\theta_r(g \otimes f)a)(m \otimes n).$$

If X also has a commuting left C-action, replacing g by cg multiplies the value by c, so θ_r preserves that action as well.

Bijectivity of θ_r. Regard M only as a right B-module. For $M = B$, evaluation at 1 identifies $\operatorname{Hom}_B(B,B) \cong B$, while multiplication identifies $B \otimes_B N \cong N$. Under these isomorphisms, θ_r becomes

$$\operatorname{Hom}_C(N,X) \otimes_B B \longrightarrow \operatorname{Hom}_C(N,X), \qquad g \otimes b \longmapsto gb,$$

which is an isomorphism by Lemma 2.2(3). Finite direct sums commute with both constructions by Lemma 2.2(4) and (2.1). Thus θ_r is bijective for $M = B^{\oplus r}$, where $r \geq 0$ is finite. If $M \in \operatorname{add} B$, choose M' with $M \oplus M' \cong B^{\oplus r}$. The comparison for $M \oplus M'$ is the direct sum of the comparisons for M and M', so both summand maps are bijective.

The left comparison. For $f \in \operatorname{Hom}_{A^{\mathrm{op}}}(M, Y)$ and $g \in \operatorname{Hom}_{B^{\mathrm{op}}}(N, B)$, well-definedness follows from

$$f(mb\, g(n)) = f(mg(bn)), \qquad f(m(gb)(n)) = f(mg(n)b) = (bf)(mg(n)) \quad (b \in B).$$

Left A-linearity of f makes the resulting functional left A-linear. For $c \in C$,

$$\theta_\ell(cg \otimes f)(m \otimes n) = f(mg(nc)) = (c\theta_\ell(g \otimes f))(m \otimes n).$$

If Y also has a commuting right A-action, replacing f by fa multiplies the value by a, so θ_ℓ preserves that action as well. For $N = B$ as a left module, evaluation at 1 identifies $\operatorname{Hom}_{B^{\mathrm{op}}}(B, B) \cong B$ and turns θ_ℓ into

$$B \otimes_B \operatorname{Hom}_{A^{\mathrm{op}}}(M, Y) \longrightarrow \operatorname{Hom}_{A^{\mathrm{op}}}(M, Y), \qquad b \otimes f \longmapsto bf.$$

The same finite-sum and direct-summand argument proves (2).

Naturality. Let $p : M \to M'$ and $q : N \to N'$ be bimodule maps. For $f \in \operatorname{Hom}_B(M', B)$ and $g \in \operatorname{Hom}_C(N', X)$, applying θ_r before or after precomposition gives $g(f(p(m))q(n))$ at $m \otimes n$. For $f \in \operatorname{Hom}_{A^{\mathrm{op}}}(M', Y)$ and $g \in \operatorname{Hom}_{B^{\mathrm{op}}}(N', B)$, the two orders give $f(p(m)g(q(n)))$ for θ_ℓ. Thus both comparisons are natural. □

Let S be finite-dimensional semisimple, and consider bimodules ${}_SV_S$, ${}_SW_S$ finitely generated on the right and left, respectively. Corollary 1.13 gives $V_S \in \operatorname{add} S$ and ${}_SW \in \operatorname{add}_{\operatorname{mod} S^{\mathrm{op}}}(S)$. Thus the semisimple case of tensor duality, Proposition 2.6, gives natural bimodule isomorphisms

$$\theta_r : W^* \otimes_S V^* \xrightarrow{\sim} (V \otimes_S W)^*, \qquad \theta_\ell : {}^*W \otimes_S {}^*V \xrightarrow{\sim} {}^*(V \otimes_S W), \tag{2.5}$$

$$\theta_r(g \otimes f)(v \otimes w) = g(f(v)w), \qquad \theta_\ell(g \otimes f)(v \otimes w) = f(vg(w)). \tag{2.6}$$

For $M \in \{V, W\}$, the actions from (1.15) are

$$(sft)(m) = sf(tm), \qquad (sgt)(m) = g(ms)t \quad \text{for all } s, t \in S,\ f \in M^*,\ g \in {}^*M,\ m \in M. \tag{2.7}$$

Example 2.7. *Finite generation alone does not suffice in Proposition 2.6 over a general algebra. Take $A = C = \Bbbk$, $B = \Bbbk[x]$, and $X = Y = \Bbbk$. Consider bimodules ${}_\Bbbk M_B$, ${}_B N_\Bbbk$ with $M = N = B/(x) \cong \Bbbk$ and the outer $\Bbbk$-actions given by scalar multiplication. Every $f \in \operatorname{Hom}_B(M, B)$ satisfies $f(\overline{1})x = f(\overline{1}x) = 0$, hence $f = 0$. Consequently*

$$\operatorname{Hom}_\Bbbk(N, \Bbbk) \otimes_B \operatorname{Hom}_B(M, B) = 0, \qquad \operatorname{Hom}_\Bbbk(M \otimes_B N, \Bbbk) \cong \Bbbk.$$

Both M_B and ${}_B N$ are generated by $\overline{1}$, so neither direct-summand hypothesis can be replaced by finite generation alone. For the left comparison, $\operatorname{Hom}_{B^{\mathrm{op}}}(N, B) = 0$, whereas $\operatorname{Hom}_\Bbbk(M \otimes_B N, \Bbbk) \cong \Bbbk$.

For S, V, W as in (2.5), take $A = B = S$, $C = \Bbbk$, $M = V$, $N = W$, and $X = \Bbbk$ in Proposition 2.6(1). Since $\operatorname{Hom}_\Bbbk(W, \Bbbk) = D(W)$, we obtain the natural bimodule isomorphism [GI20,

§2.1]

$$\theta_D : D(W) \otimes_S V^* \xrightarrow{\sim} D(V \otimes_S W),$$
$$\theta_D(\lambda \otimes f)(v \otimes w) = \lambda(f(v)w).$$

The actions on $D(W)$ are those in (1.15). For $s \in S$, naturality with respect to $W \to W$, $w \mapsto ws$, gives left S-linearity of θ_D.

Suppose now that S is equipped with a symmetrising form $\tau : S \to \Bbbk$, as in Definition 1.53. The two-sided dual of Definition 1.52 satisfies

$$\theta_{\mathrm{en}} : W^* \otimes_S V^{*\mathrm{en}} \xrightarrow{\sim} (V \otimes_S W)^{*\mathrm{en}},$$
$$\theta_{\mathrm{en}}(g \otimes F)(v \otimes w) = \sum_j g(a_j w) \otimes b_j^{\mathrm{op}}, \qquad F(v) = \sum_j a_j \otimes b_j^{\mathrm{op}}. \tag{2.8}$$

We check that the formula defines a map on the two indicated tensor products and preserves both outer actions. For example, $F(vs) = F(v)(s \otimes 1)$ replaces a_j by $a_j s$, while $(Fs)(v) = (1 \otimes s^{\mathrm{op}})F(v)$ replaces b_j by $b_j s$. These give, respectively, compatibility with $vs \otimes w = v \otimes sw$ and with the right outer action. Compatibility with $(gs) \otimes F = g \otimes (sF)$ follows from $(gs)(a_j w) = g(sa_j w)$. The other actions follow from right S-linearity of g and $F(tv) = F(v)(1 \otimes t^{\mathrm{op}})$.

The comparison maps γ_M in (1.23) give

$$\gamma_{V \otimes_S W} \circ \theta_{\mathrm{en}} = \theta_r \circ (\mathrm{id}_{W^*} \otimes \gamma_V),$$

since both sides send $g \otimes F$, evaluated at $v \otimes w$, to $\sum_j g(a_j w)\tau(b_j)$. Thus θ_{en} is an isomorphism by Theorem 1.55. Precomposition shows that θ_D and θ_{en} are natural. No assumption that S^{en} is semisimple is used; compare Remark 1.43.

For the maps $\alpha_M(f) = \tau \circ f$ in (1.18),

$$\alpha_{V \otimes_S W} \circ \theta_r = \theta_D \circ (\alpha_W \otimes \mathrm{id}_{V^*}),$$

because both sides send $g \otimes f$ to the functional $v \otimes w \mapsto \tau(g(f(v)w))$.

2.2 Generated algebras

Definition 2.8. *Let A be graded with $S = A_0$. Multiplication makes A_1 an S-bimodule and gives*

$$\mu_i : A_1^{\otimes_S i} \longrightarrow A_i, \qquad a_1 \otimes \cdots \otimes a_i \longmapsto a_1 \cdots a_i \quad (i \geq 1), \qquad \mu_0 = \mathrm{id}_S\,. \tag{2.9}$$

The algebra A is generated in degree 1 (over S) if every μ_i in (2.9) is surjective.

Equivalently, S and A_1 generate A as an algebra.

Example 2.9. *The algebra $\Bbbk A_n$ of Example 1.19 is generated in degree 1 over $S = \bigoplus_i \Bbbk e_{ii}$.*

Example 2.10. *The polynomial algebra $\Bbbk[x_1, \ldots, x_n]$, with each $\deg x_i = 1$ for all i, is generated in degree 1 over $\Bbbk$. Indeed, every monomial of total degree d is a product of d variables, and these monomials span the degree d component.*

Example 2.11. *Let*

$$A = \begin{pmatrix} \Bbbk[x] & \Bbbk[x] \\ 0 & \Bbbk[x] \end{pmatrix}, \qquad \deg(x^m e_{ij}) = m + j - i \quad (m \geq 0,\ 1 \leq i \leq j \leq 2).$$

Then

$$A_0 = \Bbbk e_{11} \oplus \Bbbk e_{22}, \qquad A_d = \Bbbk x^d e_{11} \oplus \Bbbk x^d e_{22} \oplus \Bbbk x^{d-1} e_{12} \quad (d \geq 1).$$

It is generated in degree 1 *over* A_0*, since*

$$x^d e_{11} = (xe_{11})^d, \qquad x^d e_{22} = (xe_{22})^d, \qquad x^{d-1} e_{12} = (xe_{11})^{d-1} e_{12} \quad (d \geq 1).$$

Here A_0 *has two simple summands,* $A_d \neq 0$ *for every* $d \geq 0$*, and* A *is non-commutative:* $e_{11}e_{12} = e_{12}$ *whereas* $e_{12}e_{11} = 0$*.*

2.3 Tensor algebras and the universal property

Definition 2.12. *Let* S *be a* $\Bbbk$*-algebra and consider a bimodule* ${}_S V_S$*. The tensor algebra of* V *over* S *is*

$$T_S(V) := \bigoplus_{i \in \mathbb{Z}} T_S(V)_i, \quad \text{where} \quad T_S(V)_i = \begin{cases} V^{\otimes_S i}, & i \geq 1, \\ S, & i = 0, \\ 0, & i < 0, \end{cases}.$$

For $v_1, \ldots, v_{p+q} \in V$*, multiplication is determined* $\Bbbk$*-bilinearly by*

$$(v_1 \otimes_S \cdots \otimes_S v_p)(v_{p+1} \otimes_S \cdots \otimes_S v_{p+q}) = v_1 \otimes_S \cdots \otimes_S v_{p+q} \qquad (p, q \geq 1),$$

and by the left and right S*-actions when a factor has degree* 0*.*

We also call $T_S(V)$ the tensor algebra generated by V over S. Every element has finitely many non-zero homogeneous components. The unit is 1_S, and associativity follows from Lemma 2.2(3). Write $v_1 \cdots v_d$ for $v_1 \otimes_S \cdots \otimes_S v_d$ in $T_S(V)$. In a quotient algebra (including $T_S(V)$ itself) of $T_S(V)$, the notation $v_1 \cdots v_d$ denotes the image of $v_1 \otimes_S \cdots \otimes_S v_d$.

Example 2.13 (Polynomial and exterior algebras). *Let* V, W *be finite-dimensional vector spaces with bases* $\{x_i \mid 1 \leq i \leq n\}$ *and* $\{y_j \mid 1 \leq j \leq m\}$*, respectively. The symmetric algebra of* V *and the exterior algebra of* W *are*

$$\begin{aligned} \mathsf{S}(V) &= T_\Bbbk(V)/(x_i x_j - x_j x_i \mid 1 \leq i < j \leq n) \cong \Bbbk[x_1, \ldots, x_n], \\ \bigwedge W &= T_\Bbbk(W)/(w^2 \mid w \in W). \end{aligned} \tag{2.10}$$

They are generated in degree 1*, with*

$$\mathsf{S}(V)_0 = (\textstyle\bigwedge W)_0 = \Bbbk, \qquad \mathsf{S}(V)_1 = V, \qquad (\textstyle\bigwedge W)_1 = W.$$

Their respective bases are

$$\{x_1^{a_1} \cdots x_n^{a_n} \mid a_1, \ldots, a_n \in \mathbb{Z},\ a_1, \ldots, a_n \geq 0\},$$
$$\{y_{j_1} \cdots y_{j_d} \mid 0 \leq d \leq m,\ 1 \leq j_1 < \cdots < j_d \leq m\},$$

where the empty product is 1*. The exterior relations are equivalently* $y_j^2 = 0$ *for* $1 \leq j \leq m$ *and* $y_i y_j + y_j y_i = 0$ *for* $1 \leq i < j \leq m$*, in every characteristic.*

The bases in Example 2.13 can be checked directly. For $\mathsf{S}(V)$, commuting adjacent generators gives the ordered monomials, which are independent under the map to the polynomial ring. For $\bigwedge W$, the relations reduce every product of basis elements to a multiple of a strictly increasing product or to zero. For $1 \le d \le m$, fix $j_1 < \cdots < j_d$ and consider the linear functional $W^{\otimes_{\Bbbk} d} \to \Bbbk$ sending $w_1 \otimes \cdots \otimes w_d$ to the determinant of their coordinates in rows $j_1, \dots, j_d$ relative to the basis $\{y_j \mid 1 \le j \le m\}$. The determinant vanishes whenever two adjacent factors agree, hence on the degree d component of $(w^2 \mid w \in W)$. On increasing products it is 1 on $y_{j_1} \cdots y_{j_d}$ and 0 on the others, proving independence in every characteristic.

Lemma 2.14. *Let $S = \bigoplus_{i=1}^{r} \Bbbk e_i$ with coordinate idempotents, and consider bimodules ${}_S V_S$, ${}_S W_S$. There is a bimodule isomorphism*

$$V \otimes_S W \cong \bigoplus_{i,j,k=1}^{r} e_i V e_j \otimes_{\Bbbk} e_j W e_k. \tag{2.11}$$

Consequently, after choosing a basis of each $e_i V e_j$, a basis of $V^{\otimes_S d}$ for $d \ge 1$ consists of products of d chosen basis elements whose adjacent idempotents match.

Proof The orthogonal idempotents sum to 1, so

$$V = \bigoplus_{i,j=1}^{r} e_i V e_j, \qquad (e_i V e_j) \otimes_S (e_p W e_k) = 0 \quad (1 \le i, j, p, k \le r,\ j \ne p). \tag{2.12}$$

Indeed, $v \otimes_S w = v e_j \otimes_S w = v \otimes_S e_j w = 0$ for $v \in e_i V e_j$, $w \in e_p W e_k$, and $j \ne p$. The map from right to left in (2.11) sends $v \otimes_{\Bbbk} w$ to $v \otimes_S w$ on each summand. Its inverse is

$$V \otimes_S W \longrightarrow \bigoplus_{i,j,k=1}^{r} e_i V e_j \otimes_{\Bbbk} e_j W e_k, \qquad v \otimes_S w \longmapsto \sum_{i,j,k=1}^{r} e_i v e_j \otimes_{\Bbbk} e_j w e_k.$$

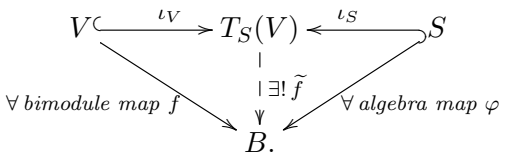
Checking that these two assignments induce well-defined, mutually inverse maps is left to the reader. Iterating (2.11) and using Definition 1.36 gives the stated basis. □

Example 2.15. *We apply Lemma 2.14 to realise $\Bbbk A_3$ as a tensor algebra. Let $S = \Bbbk^{\oplus 3}$ with coordinate idempotents e_1, e_2, e_3, and take the bimodule ${}_S V_S$ with $V = \Bbbk a \oplus \Bbbk b$, $e_1 a e_2 = a$, and $e_2 b e_3 = b$; all other idempotent actions are zero. By Lemma 2.14,*

$$V \otimes_S V = \Bbbk(a \otimes b), \qquad V^{\otimes_S 3} = 0, \qquad T_S(V) = S \oplus \Bbbk a \oplus \Bbbk b \oplus \Bbbk ab.$$

Here $ab = a \otimes b$. The linear map $T_S(V) \to \Bbbk A_3$ given by $e_i \mapsto e_{ii}$, $a \mapsto e_{12}$, $b \mapsto e_{23}$, and $ab \mapsto e_{13}$ preserves multiplication and sends a basis to a basis, hence is an algebra isomorphism.

Proposition 2.16 (Universal property of the tensor algebra). *Let S be a $\Bbbk$-algebra, consider a bimodule ${}_S V_S$, and let $\iota_S : S \to T_S(V)$ and $\iota_V : V \to T_S(V)$ be the canonical inclusions. Given a unital $\Bbbk$-algebra homomorphism $\varphi : S \to B$ and a bimodule map $f : {}_S V_S \to {}_S B_S$, where $sbt = \varphi(s) b \varphi(t)$ on B, there is a unique $\Bbbk$-algebra homomorphism $\widetilde{f} : T_S(V) \to B$ making the diagram commute:*



(2.13)

Proof Define $\widetilde{f}$ on S and each tensor power by

$$\widetilde{f}(s) = \varphi(s), \qquad \widetilde{f}(v_1 \otimes_S \cdots \otimes_S v_i) = f(v_1)\cdots f(v_i) \quad (s \in S,\ i \geq 1). \tag{2.14}$$

The identity $f(vs)f(w) = f(v)\varphi(s)f(w) = f(v)f(sw)$ shows that replacing adjacent factors $vs \otimes w$ by $v \otimes sw$ does not change the value in (2.14). The formulas therefore induce linear maps on the tensor powers, which extend to their direct sum. Multiplication is preserved by (2.14), including the degree 0 factors because $f(sv) = \varphi(s)f(v)$ and $f(vs) = f(v)\varphi(s)$. Also, $\widetilde{f}(1_S) = 1_B$. Every algebra homomorphism making (2.13) commute must have the values in (2.14), proving uniqueness. □

If B is graded, $\varphi(S) \subset B_0$, and $f(V) \subset B_1$, then (2.14) gives $\widetilde{f}(V^{\otimes_S i}) \subset B_i$ for every $i \geq 0$. Thus $\widetilde{f}$ is a graded algebra homomorphism.

Corollary 2.17. *If A is generated in degree 1 over $S = A_0$, multiplication gives a surjective graded algebra homomorphism*

$$\mu : T_S(A_1) \twoheadrightarrow A, \qquad a_1 \otimes_S \cdots \otimes_S a_i \longmapsto a_1 \cdots a_i \quad (i \geq 1), \qquad \mu(s) = s \quad (s \in S).$$

Its degree i component is μ_i from (2.9).

Proof In Proposition 2.16, take $V = A_1$, $B = A$, and the inclusions

$$\varphi : S \hookrightarrow A, \quad s \longmapsto s, \qquad f : A_1 \hookrightarrow A, \quad a \longmapsto a.$$

Then (2.14) gives $\widetilde{f} = \mu$. The map μ preserves degrees, and each component μ_i is surjective by the assumption that A is generated in degree 1. □

Definition 2.18. *For the map μ in Corollary 2.17, the homogeneous two-sided ideal $I := \operatorname{Ker} \mu$ is the ideal of relations.*

Writing $V = A_1$, the components $\mu_0 = \mathrm{id}_S$ and $\mu_1 = \mathrm{id}_V$ give $I_0 = I_1 = 0$. Since μ preserves degrees, its kernel is homogeneous, and

$$T_S(V)/I \xrightarrow{\sim} A, \qquad z + I \longmapsto \mu(z)$$

is a graded algebra isomorphism. The bimodule V specifies the generators and their left and right S-actions; I specifies the relations between their products. Homogeneity is essential for the quotient grading: the relation $x - x^2$ in Example 1.34 does not satisfy this condition.

2.4 Quadratic algebras

We now consider algebras whose ideal of relations is generated in degree 2.

Definition 2.19. *A graded algebra A is quadratic if there exist a finite-dimensional semisimple $\Bbbk$-algebra S, a finite-dimensional S-bimodule V, and an S-sub-bimodule $R \subset V \otimes_S V$ such that $A \cong T_S(V)/(R)$ as graded algebras, where V is in degree 1 and (R) is the two-sided ideal of $T_S(V)$ generated by R.*

An isomorphism $A \cong T_S(V)/(R)$ as in Definition 2.19 is called a quadratic presentation of A.

Proposition 2.20. *A quadratic algebra $A = T_S(V)/(R)$ has $A_0 = S$, $A_1 = V$, and $A_2 = (V \otimes_S V)/R$, is locally finite, and is generated in degree 1 over S.*

Proof The ideal (R) has no components in degrees 0 and 1. Its degree 2 component is exactly R. Each A_i is a quotient of the finite-dimensional space $V^{\otimes_S i}$, and multiplication maps $V^{\otimes_S i}$ onto A_i. □

Thus a quadratic presentation recovers $S \cong A_0$, $V \cong A_1$, and $R \cong \operatorname{Ker}(A_1 \otimes_{A_0} A_1 \to A_2)$. In particular, the kernel of the canonical map $T_{A_0}(A_1) \twoheadrightarrow A$ is generated by its degree 2 component. Quadraticity depends on the grading: for $n \geq 2$, the grading $\deg e_{ij} = 2(j-i)$ on $\Bbbk A_n$ is not quadratic, since it has no degree 1 generators.

Example 2.21. *Let $S = \bigoplus_{i=1}^{n} \Bbbk e_{ii}$ and $V = \bigoplus_{i=1}^{n-1} \Bbbk e_{i,i+1}$ inside $\Bbbk A_n$. By Lemma 2.14, the basis tensors of positive length are*

$$e_{i,i+1} \otimes_S e_{i+1,i+2} \otimes_S \cdots \otimes_S e_{j-1,j} \longmapsto e_{ij} \quad (1 \leq i < j \leq n).$$

Thus multiplication gives $T_S(V) \cong \Bbbk A_n$, a quadratic algebra with $R = 0$.

Example 2.22. *For S, V, a, b from Example 2.15, the quotient*

$$T_S(V)/(ab) \cong \Bbbk A_3/(e_{13})$$

is quadratic with $R = \Bbbk(a \otimes b)$. Its basis is e_1, e_2, e_3, a, b, in degrees $0, 0, 0, 1, 1$, as in Example 1.33.

Example 2.23. *Let $S = \Bbbk^{\oplus 2}$ with coordinate idempotents e_1, e_2, and let $V = \Bbbk a \oplus \Bbbk b$ satisfy $e_1 a e_2 = a$ and $e_2 b e_1 = b$, with all other idempotent actions zero. By Lemma 2.14, the alternating words in a, b form the basis of positive degree tensors in $T_S(V)$. The quadratic algebra $A = T_S(V)/(ab)$ has*

$$A = S \oplus \Bbbk a \oplus \Bbbk b \oplus \Bbbk ba, \qquad \dim_{\Bbbk} A_0 = 2, \quad \dim_{\Bbbk} A_1 = 2, \quad \dim_{\Bbbk} A_2 = 1.$$

Every alternating word of length at least 3 contains ab, so all higher components vanish.

Example 2.24. *Let $B = \Bbbk A_4$ and $A = B/(e_{14})$, with the grading of Example 1.19. The ideal $I = (e_{14}) = \Bbbk e_{14}$ has $I_2 = 0$ but $I_3 \neq 0$. Since $e_{14} = e_{12} e_{23} e_{34}$, the algebra A is generated in degree 1 with a relation of degree 3 and no relations of degree 2. By Proposition 2.20, any quadratic presentation would have the same generators and zero quadratic relation space, hence would give $\Bbbk A_4$ itself. Thus A is not quadratic.*

Example 2.25. *The symmetric and exterior algebras of Example 2.13 are quadratic: the defining relations in* (2.10) *belong to $V \otimes_{\Bbbk} V$ and $W \otimes_{\Bbbk} W$, respectively.*

2.5 Hilbert series

Fix a finite-dimensional semisimple $\Bbbk$-algebra S, concentrated in degree 0. We record dimensions relative to all simple right S-modules. We define module series first, then collect them in an array for a graded algebra.

Definition 2.26. *Choose representatives $L_1, \ldots, L_r$ of the isomorphism classes of simple right S-modules. For a finite-dimensional right S-module M, write, with $m_j \in \mathbb{Z}_{\geq 0}$,*

$$M \cong \bigoplus_{j=1}^{r} L_j^{\oplus m_j}, \qquad \mathbf{dim} M := (m_1, \ldots, m_r).$$

The row vector $\mathbf{dim} M$ is the dimension vector of M. For a finite-dimensional bimodule ${}_S U_S$, arrange the vectors $\mathbf{dim}(L_i \otimes_S U)$ as rows:

$$\mathbf{dim} U = (u_{ij})_{1 \leq i,j \leq r}, \qquad (\mathbf{dim} U)_{ij} := \big(\mathbf{dim}(L_i \otimes_S U)\big)_j.$$

Thus $L_i \otimes_S U \cong \bigoplus_{j=1}^{r} L_j^{\oplus u_{ij}}$.

The decompositions exist by Theorem 1.12. Their multiplicities are unique: a non-zero homomorphism between simple modules is an isomorphism, since its kernel and image are submodules. Consequently

$$m_j = \frac{\dim_{\Bbbk} \operatorname{Hom}_S(L_j, M)}{\dim_{\Bbbk} \operatorname{End}_S(L_j)},$$

and the same formula applies to each $L_i \otimes_S U$. In particular, the dimension data do not depend on choices of decompositions. For the S-bimodule S, we have $\mathbf{dim} S = I_r$, the $r \times r$ identity array.

Example 2.27. *If $S = \bigoplus_{i=1}^{r} \Bbbk e_i$ with coordinate idempotents, take $L_i = e_i S$. Then*

$$\mathbf{dim} M = (\dim_{\Bbbk} M e_j)_{j=1}^{r}, \qquad (\mathbf{dim} U)_{ij} = \dim_{\Bbbk}(e_i U e_j).$$

Indeed, $L_i \otimes_S U \cong e_i U$ by multiplication, and the multiplicity of $e_j S$ in $e_i U$ is $\dim_{\Bbbk}(e_i U e_j)$. For $S = \Bbbk$, the dimension vector has one entry, the usual vector space dimension.

Definition 2.28. *Let $M = \bigoplus_{d \in \mathbb{Z}} M_d$ be a locally finite graded right S-module, with $M_d = 0$ for all sufficiently negative d. Its Hilbert series is*

$$H_M(t) := \sum_{d \in \mathbb{Z}} (\mathbf{dim} M_d) t^d.$$

For a locally finite graded algebra A with $A_0 = S$, its Hilbert series records the preceding module series row by row:

$$H_A(t)_{ij} := \big(H_{L_i \otimes_S A}(t)\big)_j \quad (1 \le i, j \le r), \qquad H_A(t) = \sum_{d \ge 0} (\mathbf{dim} A_d) t^d.$$

Here each L_i is placed in degree 0, and $L_i \otimes_S A$ is regarded as a right S-module by restriction. In particular, $H_A(0) = I_r$. For $S = \bigoplus_{i=1}^{r} \Bbbk e_i$, this gives

$$H_A(t)_{ij} = \sum_{d \ge 0} \dim_{\Bbbk}(e_i A_d e_j) t^d \quad (1 \le i, j \le r).$$

The entries are formal Laurent series with integer coefficients; for algebras they are formal power series. A finite-dimensional graded module or algebra gives a Laurent polynomial or polynomial, respectively. No numerical convergence is required: a coefficient is specified degree by degree. For an algebra, keeping both indices records where products can be formed.

Lemma 2.29. *For locally finite graded right S-modules bounded below:*

(1) *A graded isomorphism $M \cong N$ implies $H_M(t) = H_N(t)$.*

(2) *A short exact sequence $0 \to L \to M \to N \to 0$ gives $H_M(t) = H_L(t) + H_N(t)$.*

(3) *For every $a \in \mathbb{Z}$, $H_{M(a)}(t) = t^{-a} H_M(t)$.*

Proof A graded isomorphism preserves each degree and its simple-module multiplicities. A short exact sequence of right S-modules splits by semisimplicity, so multiplicities add. Finally, $M(a)_d = M_{d+a}$ gives

$$H_{M(a)}(t) = \sum_d (\mathbf{dim} M_{d+a}) t^d = t^{-a} \sum_j (\mathbf{dim} M_j) t^j. \qquad \square$$

Definition 2.30. *For graded right and left S-modules M, N, their tensor product has total grading*

$$(M \otimes_S N)_d = \bigoplus_{p+q=d} M_p \otimes_S N_q.$$

The same definition applies to bimodules, and to vector spaces by taking $S = \Bbbk$.

The decomposition follows by distributing tensor products over direct sums, as in Lemma 2.2(4). When M, N are locally finite and bounded below, only finitely many pairs (p, q) contribute in a fixed degree, so the tensor product is locally finite and bounded below.

To multiply two arrays $C = (c_{ij})$ and $D = (d_{ij})$, or a row vector $m = (m_i)$ and an array C, use

$$(CD)_{i\ell} = \sum_{j=1}^{r} c_{ij} d_{j\ell}, \qquad (mC)_j = \sum_{i=1}^{r} m_i c_{ij}.$$

Thus multiplication sums over the matching middle index, rather than multiplying individual entries independently. For series, use the same products on coefficients and collect terms with the same total power of t.

Lemma 2.31. *Let ${}_SU_S$, ${}_SW_S$ be finite-dimensional bimodules and M a finite-dimensional right S-module. Then*

$$\mathbf{dim}(U \otimes_S W) = (\mathbf{dim} U)(\mathbf{dim} W), \qquad \mathbf{dim}(M \otimes_S U) = (\mathbf{dim} M)(\mathbf{dim} U).$$

For a locally finite graded right S-module M bounded below and a locally finite graded algebra A with $A_0 = S$, this gives

$$H_{M \otimes_S A}(t) = H_M(t) H_A(t).$$

Proof Write $C = \mathbf{dim} U$ and $D = \mathbf{dim} W$. Associativity and distribution over direct sums give

$$\begin{aligned} L_i \otimes_S (U \otimes_S W) &\cong (L_i \otimes_S U) \otimes_S W \\ &\cong \bigoplus_j (L_j \otimes_S W)^{\oplus c_{ij}} \cong \bigoplus_\ell L_\ell^{\oplus \sum_j c_{ij} d_{j\ell}}. \end{aligned}$$

This proves the first equality. Decomposing $M \cong \bigoplus_i L_i^{\oplus m_i}$ proves the second in the same way. Apply the second equality to each $M_p \otimes_S A_q$ and sum over $p + q = d$ to obtain the series identity. □

Example 2.32 (Hilbert series of a tensor algebra). *Let ${}_SV_S$ be a finite-dimensional bimodule, placed in degree 1, and put $C = \mathbf{dim} V$. Since $T_S(V)_d = V^{\otimes_S d}$ and $V^{\otimes_S 0} = S$, Lemma 2.31 gives*

$$\mathbf{dim} T_S(V)_d = C^d \quad (d \geq 0), \qquad C^0 = I_r.$$

Consequently,

$$\boxed{H_{T_S(V)}(t) = \sum_{d \geq 0} t^d C^d = (I_r - tC)^{-1} = (I_r - t\,\mathbf{dim} V)^{-1}.} \tag{2.15}$$

Indeed, multiplying the series on either side by $I_r - tC$ gives constant coefficient I_r; the coefficient of t^d for $d \geq 1$ is $C^d - C^d = 0$. The inverse is therefore an identity of formal power series with the coefficient multiplication defined above.

For $S = \Bbbk$ and $\dim_\Bbbk V = m$, the formula is $H_{T_\Bbbk(V)}(t) = 1/(1 - mt)$. More generally, for $S = \mathrm{Mat}_n(\Bbbk)$ and $V = S^{\oplus m}$ as an S-bimodule, there is one simple right S-module $L = \Bbbk^{1 \times n}$. Since $L \otimes_S V \cong L^{\oplus m}$, we again have $C = (m)$ and $H_{T_S(V)}(t) = 1/(1 - mt)$. Here the coefficient records simple-module multiplicities, not $\dim_\Bbbk V = mn^2$.

Example 2.33 (Upper triangular algebras and a quotient). *For $A = \Bbbk A_n$, order the coordinate idempotents as $e_{11}, \ldots, e_{nn}$. Let C have $C_{i,i+1} = 1$ for $1 \leq i < n$ and all other entries zero. The tensor presentation in Example 2.21 and* (2.15) *give*

$$H_A(t) = I_n + tC + \cdots + t^{n-1} C^{n-1}, \qquad H_A(t)_{ij} = \begin{cases} t^{j-i}, & i \leq j, \\ 0, & i > j. \end{cases}$$

For $n = 3$, killing e_{13} removes the degree 2 contribution in entry $(1,3)$:

$$H_{\Bbbk A_3}(t) = \begin{pmatrix} 1 & t & t^2 \\ 0 & 1 & t \\ 0 & 0 & 1 \end{pmatrix}, \qquad H_{\Bbbk A_3/(e_{13})}(t) = \begin{pmatrix} 1 & t & 0 \\ 0 & 1 & t \\ 0 & 0 & 1 \end{pmatrix}.$$

The quotient removes $\Bbbk e_{13}$ from degree 2, and hence removes t^2 from entry $(1,3)$.

Example 2.34 (Two alternating generators). *For S, V, a, b in Example 2.23, use the order e_1, e_2. Then*

$$C = \mathbf{dim} V = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad C^2 = I_2, \qquad H_{T_S(V)}(t) = \frac{1}{1-t^2}\begin{pmatrix} 1 & t \\ t & 1 \end{pmatrix}.$$

In $A = T_S(V)/(ab)$, the degree 2 basis vector ba belongs to $e_2 A e_2$, so

$$H_A(t) = \begin{pmatrix} 1 & t \\ t & 1+t^2 \end{pmatrix}.$$

Example 2.35. *For the algebras in Example 2.13, $S = \Bbbk$ and there is one dimension entry. Each polynomial generator may occur any number of times, whereas each exterior generator occurs at most once. Thus*

$$H_{\mathsf{S}(V)}(t) = \frac{1}{(1-t)^n}, \qquad H_{\bigwedge W}(t) = (1+t)^m \quad \text{where } n = \dim_{\Bbbk} V,\ m = \dim_{\Bbbk} W.$$

Lemma 2.36. *For locally finite graded vector spaces U, W bounded below,*

$$H_{U \otimes_{\Bbbk} W}(t) = H_U(t) H_W(t).$$

Let A, B be locally finite graded algebras with $A_0 = \Bbbk^{\oplus r}$ and $B_0 = \Bbbk^{\oplus s}$. Write $e_1, \dots, e_r$ and $f_1, \dots, f_s$ for the coordinate idempotents. Give $A \otimes_{\Bbbk} B$ the total grading and multiplication $(a \otimes b)(a' \otimes b') = aa' \otimes bb'$. Then

$$H_{A \otimes_{\Bbbk} B}(t)_{(i,p),(j,q)} = H_A(t)_{ij} H_B(t)_{pq} \quad (1 \le i, j \le r,\ 1 \le p, q \le s).$$

Proof For vector spaces, take dimensions in each summand of the total grading. For algebras,

$$(e_i \otimes f_p)(A_a \otimes_{\Bbbk} B_b)(e_j \otimes f_q) = (e_i A_a e_j) \otimes_{\Bbbk} (f_p B_b f_q).$$

Take vector space dimensions and sum over $a + b = d$ to obtain the coefficient of t^d. □

2.6 The degree 2 relations of a quotient

Let S be finite-dimensional semisimple, V a finite-dimensional S-bimodule, and $I \subset T_S(V)$ a homogeneous two-sided ideal with $I_0 = I_1 = 0$. Set $A = T_S(V)/I$. In degree 2,

$$A_2 \cong (V \otimes_S V)/I_2, \quad \text{where} \quad I_2 = I \cap (V \otimes_S V).$$

Definition 2.37. *The quadratic part of A is $qA := T_S(V)/(I_2)$.*

Since $(I_2) \subset I$, there is a surjective graded algebra homomorphism

$$\pi_A : qA \longrightarrow A, \qquad z + (I_2) \longmapsto z + I, \qquad \operatorname{Ker} \pi_A = I/(I_2).$$

Consequently,

$$\pi_A \text{ is an isomorphism} \iff I = (I_2) \iff A \text{ is quadratic.}$$

For the last equivalence, (I_2) gives a quadratic presentation, while Proposition 2.20 identifies the generators and degree 2 relations of any quadratic presentation with A_1 and I_2.

Example 2.38. *For $A = \Bbbk A_4/(e_{14})$ from Example 2.24, we have $I_2 = 0$, so*

$$qA = \Bbbk A_4, \qquad \pi_A : \Bbbk A_4 \twoheadrightarrow \Bbbk A_4/(e_{14}), \qquad \operatorname{Ker} \pi_A = \Bbbk e_{14}.$$

Thus qA does not impose the cubic relation $e_{12}e_{23}e_{34} = 0$.

2.7 Orthogonal complements and quadratic duality

Let S be finite-dimensional semisimple.

Definition 2.39. *For an S-sub-bimodule U of a finite-dimensional S-bimodule M, its right orthogonal complement and left orthogonal complement are*

$$U^{\perp} := \{f \in M^* \mid f(u) = 0 \text{ for all } u \in U\},$$
$$^{\perp}U := \{g \in {}^*M \mid g(u) = 0 \text{ for all } u \in U\}.$$

Both orthogonal complements are S-sub-bimodules. Indeed, for $s, t \in S$ and $u \in U$, (2.7) gives

$$(sft)(u) = sf(tu) = 0 \quad (f \in U^{\perp}), \qquad (sgt)(u) = g(us)t = 0 \quad (g \in {}^{\perp}U),$$

since $tu, us \in U$. The orthogonal complements are defined without choosing a symmetrising form.

If S is equipped with a symmetrising form $\tau : S \to \Bbbk$, the comparison maps α_M, β_M of (1.18) satisfy

$$\alpha_M(U^{\perp}) = \beta_M({}^{\perp}U) = \{\lambda \in D(M) \mid \lambda(u) = 0 \text{ for all } u \in U\}. \tag{2.16}$$

The forward inclusions follow by composing with τ. Conversely, $\alpha_M(f)(U) = 0$ gives

$$\tau(f(u)s) = \tau(f(us)) = 0 \quad \text{for all } s \in S,\ u \in U.$$

Non-degeneracy of τ forces $f(u) = 0$ for every $u \in U$. For the left dual, $\tau(sg(u)) = \tau(g(su)) = 0$ gives $g(u) = 0$ instead. Bijectivity of α_M, β_M in Theorem 1.55 proves (2.16).

Example 2.40. *We compute the orthogonal complements of the symmetric and exterior relation spaces separately. Take $V = \Bbbk x_1 \oplus \Bbbk x_2$ and $W = \Bbbk y_1 \oplus \Bbbk y_2$ in Example 2.13. The relation spaces in $V \otimes_{\Bbbk} V$ and $W \otimes_{\Bbbk} W$, respectively, are*

$$R_{\mathsf{S}(V)} = \Bbbk(x_1 \otimes x_2 - x_2 \otimes x_1),$$
$$R_{\bigwedge W} = \Bbbk(y_1 \otimes y_1) \oplus \Bbbk(y_2 \otimes y_2) \oplus \Bbbk(y_1 \otimes y_2 + y_2 \otimes y_1).$$

Let x_1^*, x_2^* and y_1^*, y_2^* be the dual bases of $D(V)$ and $D(W)$, respectively. Under the tensor-dual identification (2.6),

$$R_{\mathsf{S}(V)}^{\perp} = \operatorname{span}_{\Bbbk}\{x_1^* \otimes x_1^*,\ x_2^* \otimes x_2^*,\ x_1^* \otimes x_2^* + x_2^* \otimes x_1^*\},$$
$$R_{\bigwedge W}^{\perp} = \Bbbk(y_1^* \otimes y_2^* - y_2^* \otimes y_1^*).$$

The first space consists of the exterior relations on $D(V)$; the second consists of the symmetric relations on $D(W)$.

2.8 Exercises

Exercise 2.41 (Interaction of tensor product with S-dual). *Let S be finite-dimensional semisimple, and consider finite-dimensional bimodules ${}_SV_S$, ${}_SW_S$. Prove the right S-dual isomorphism* (2.5)*, the semisimple case of Proposition 2.6.*

(1) *Check that the formula for θ_r in* (2.6) *is bilinear in each pair of variables, is unchanged by*

$$vs \otimes w \longleftrightarrow v \otimes sw, \qquad gs \otimes f \longleftrightarrow g \otimes sf \quad (s \in S),$$

and gives a right S-linear map $V \otimes_S W \to S$ for each $g \otimes f$. Verify $\theta_r(sg \otimes ft) = s\theta_r(g \otimes f)t$ for $s, t \in S$.

(2) *Prove that θ_r is bijective.*

(3) *For bimodule maps $p : V \to V'$ and $q : W \to W'$, prove*

$$(p \otimes q)^* \circ \theta_r^{V',W'} = \theta_r^{V,W} \circ (q^* \otimes p^*).$$

Here the superscripts specify the pair of bimodules, and p^*, q^*, and $(p \otimes q)^*$ are given by precomposition with p, q, and $p \otimes q$, respectively.

Hint for (2). *First take $V = S$, then use Corollary 1.13.*

Remark. *The analogous assertions hold for left S-duals, with left and right interchanged.*

Exercise 2.42. *Let $A = \Bbbk A_4/(e_{13}, e_{24})$ with the grading of Example 1.19. Find a homogeneous basis, compute $H_A(t)$, and write $A = T_S(V)/(R)$ with explicit S, V, R.* Hint. *Use Lemma 2.14; the ideal contains e_{14} as well as e_{13}, e_{24}.*

Exercise 2.43. *For the bimodule V in Example 2.23, find a basis of each $T_S(V)_d$ and show that*

$$H_{T_S(V)}(t) = \frac{1}{1-t^2}\begin{pmatrix} 1 & t \\ t & 1 \end{pmatrix}.$$

Compare with the quotient $T_S(V)/(ab)$. Hint. *Count alternating words separately in each degree, including degree 0.*

Exercise 2.44. *For the algebras in Example 2.13, prove*

$$\dim_{\Bbbk} \mathsf{S}(V)_2 = \frac{n(n+1)}{2}, \qquad \dim_{\Bbbk}(\bigwedge W)_2 = \frac{m(m-1)}{2} \quad \text{where } n = \dim_{\Bbbk} V,\ m = \dim_{\Bbbk} W.$$

Hint. *Count the ordered pairs allowed in the two bases of Example 2.13.*

3 The quadratic dual

3.1 Generators and orthogonal relations

Fix a quadratic presentation $A = T_S(V)/(R)$ as in Definition 2.19: S is finite-dimensional semisimple, V is a finite-dimensional S-bimodule, and $R \subset V \otimes_S V$ is an S-sub-bimodule. We will replace the generators by an S-dual and the relations by the S-linear maps that vanish on R. Use $V^* = \mathrm{Hom}_S(V,S)$ and ${}^*V = \mathrm{Hom}_{S^{\mathrm{op}}}(V,S)$ from Definition 1.49, with the bimodule actions in (2.7). The tensor isomorphisms (2.5) have values

$$\begin{aligned} \theta_r(g \otimes f)(v \otimes w) &= g(f(v)w), \quad f,g \in V^*, \\ \theta_\ell(g \otimes f)(v \otimes w) &= f(vg(w)), \quad f,g \in {}^*V. \end{aligned} \tag{3.1}$$

Here $v, w \in V$. Both isomorphisms reverse the order of the factors.

Convention. *We identify $V^* \otimes_S V^*$ with $(V \otimes_S V)^*$ via θ_r, and ${}^*V \otimes_S {}^*V$ with ${}^*(V \otimes_S V)$ via θ_ℓ. Thus $R^\perp$ and ${}^\perp R$ in tensor algebras mean the inverse images of the orthogonal complements in Definition 2.39 under these evaluations.*

Definition 3.1. *The right quadratic dual and left quadratic dual of A are*

$$\begin{aligned} A^! &:= T_S(V^*)/(R^\perp), \quad R^\perp = \{F \in V^* \otimes_S V^* \mid \theta_r(F)(R) = 0\}, \\ {}^!A &:= T_S({}^*V)/({}^\perp R), \quad {}^\perp R = \{F \in {}^*V \otimes_S {}^*V \mid \theta_\ell(F)(R) = 0\}. \end{aligned} \tag{3.2}$$

We use $A^!$ unless a side is specified. The ordinary S-duals V^* and ${}^*V$ are placed in degree 1 in (3.2). By contrast, when V has degree 1, its graded S-dual $V^{(*)}$ has degree -1 by Definition 1.50.

By Proposition 2.20, a quadratic presentation identifies S with A_0, V with A_1, and R with $\operatorname{Ker}\mu_2$, where $\mu_2 : A_1 \otimes_{A_0} A_1 \to A_2$ is (2.9). The natural tensor-dual isomorphisms (2.5) therefore make the dual algebras independent of the chosen presentation up to isomorphism. An isomorphism *over S* means one restricting to id_S on degree 0.

> The constructions are those of [BGS96, §§2.7–2.8], with different star conventions: their V^* uses left S-linear maps and their ${}^*V$ uses right S-linear maps. Thus our $A^!$ is their ${}^!A$, and our ${}^!A$ is their $A^!$.

Proposition 3.2. *The algebra $A^!$ is quadratic and*

$$A^!_0 = S, \qquad A^!_1 = V^*, \qquad A^!_2 \cong R^*.$$

In particular, $\dim_{\Bbbk} A^!_2 = \dim_{\Bbbk} R$.

Proof Restriction to R gives an exact sequence of S-bimodules

$$0 \longrightarrow R^\perp \longrightarrow V^* \otimes_S V^* \xrightarrow{F \mapsto \theta_r(F)|_R} R^* \longrightarrow 0. \tag{3.3}$$

Surjectivity follows from Theorem 1.12: R has a complement as a right S-module, so a right S-linear map $R \to S$ extends by zero on that complement. Taking the quotient in (3.3) gives $A^!_2 \cong R^*$; Proposition 2.20 gives the remaining assertions about $A^!$.

> For the dimension assertion, Theorem 1.12 reduces to simple right modules over matrix rings. For a finite-dimensional division algebra Δ, the right $\mathrm{Mat}_n(\Delta)$-module $\Delta^{1\times n}$ has dual $\Delta^{n\times 1}$: a column c gives the homomorphism $x \mapsto cx$. The row and column spaces have the same $\Bbbk$-dimension, and dimensions add over simple summands. Hence $\dim_{\Bbbk} R^* = \dim_{\Bbbk} R$ without choosing a symmetrising form. $\square$

Adding relations has the opposite effect after dualising. For S-sub-bimodules $R \subset R' \subset V \otimes_S V$,

$$R \subset R' \quad \Longrightarrow \quad (R')^{\perp} \subset R^{\perp} \quad \Longrightarrow \quad \big(T_S(V)/(R')\big)^! \twoheadrightarrow \big(T_S(V)/(R)\big)^!.$$

The displayed surjection fixes S and is induced by id_{V^*} on the generators.

3.2 Recovering the original algebra

Lemma 3.3. *Let $U \subset M$ be an S-sub-bimodule of a finite-dimensional S-bimodule. Under the evaluation isomorphisms of Lemma 1.51,*

$$^{\perp}(U^{\perp}) = U, \qquad (^{\perp}U)^{\perp} = U.$$

Proof For $u \in U$ and $f \in U^{\perp}$, evaluation gives $\mathrm{ev}_M(u)(f) = f(u) = 0$, so $U \subset {}^{\perp}(U^{\perp})$ under Lemma 1.51. If $m \notin U$, Corollary 1.13 embeds M/U in $S^{\oplus n}$ as a right module for some n. A coordinate of the image of $m + U$ is non-zero. Composing $M \twoheadrightarrow M/U \hookrightarrow S^{\oplus n}$ with that coordinate projection gives a right S-linear map $f : M \to S$ satisfying $f(U) = 0$ and $f(m) \neq 0$. Thus $f \in U^{\perp}$ and $m \notin {}^{\perp}(U^{\perp})$. The proof of $(^{\perp}U)^{\perp} = U$ uses left modules instead. □

Theorem 3.4. *Left and right quadratic duality are canonically inverse:*

$$^!(A^!) \cong A \cong (^!A)^! \quad \textit{as graded algebras over } S.$$

Proof *Generators.* The evaluation $\mathrm{ev}_V : V \xrightarrow{\sim} {}^*(V^*)$, $v \mapsto (f \mapsto f(v))$, of Lemma 1.51 identifies the generators of A and $^!(A^!)$. It preserves both S-actions by (2.7).

Relations. For $v, w \in V$ and $f, g \in V^*$, the same actions and (3.1) give

$$\begin{aligned} \theta_\ell(\mathrm{ev}_V(v) \otimes \mathrm{ev}_V(w))(g \otimes f) &= \mathrm{ev}_V(w)(gf(v)) = g(f(v)w) \\ &= \mathrm{ev}_{V \otimes_S V}(v \otimes w)(\theta_r(g \otimes f)). \end{aligned} \tag{3.4}$$

Thus $\mathrm{ev}_V \otimes \mathrm{ev}_V$ agrees with evaluation on $V \otimes_S V$ under the tensor-dual identifications (2.5). Lemma 3.3, applied to $R \subset V \otimes_S V$, identifies the relations ${}^{\perp}(R^{\perp})$ of $^!(A^!)$ with R. The presentations (3.2) therefore give $^!(A^!) \cong A$ over S. Interchanging left and right gives $(^!A)^! \cong A$.

□

Proposition 3.5. *A choice of symmetrising form $\tau : S \to \Bbbk$ gives isomorphisms $A^! \cong {}^!A$ and $(A^!)^! \cong A$ of graded algebras over S.*

Proof The maps $\alpha_V(f) = \tau \circ f$ and $\beta_V(g) = \tau \circ g$ in (1.18) give a bimodule isomorphism $\beta_V^{-1}\alpha_V : V^* \to {}^*V$. For $f, g \in V^*$, write $f' = \beta_V^{-1}\alpha_V(f)$ and $g' = \beta_V^{-1}\alpha_V(g)$. Then

$$\begin{aligned} \tau\big(f'(vg'(w))\big) &= \tau\big(f(v)g'(w)\big) = \tau\big(g'(f(v)w)\big) \\ &= \tau\big(g(f(v)w)\big) \qquad (v, w \in V). \end{aligned}$$

By (2.16), $(\beta_V^{-1}\alpha_V)^{\otimes_S 2}$ therefore identifies $R^{\perp}$ with ${}^{\perp}R$. The bimodule map $\beta_V^{-1} \circ \alpha_V$ extends by Proposition 2.16 to an isomorphism $T_S(V^*) \to T_S({}^*V)$ and hence descends to $A^! \cong {}^!A$. Apply the same comparison to $A^!$ and then Theorem 3.4 to obtain $(A^!)^! \cong {}^!(A^!) \cong A$. □

Theorem 3.4 needs no choice of form. Proposition 3.5 permits the usual statement that quadratic duality is an involution, but the identification of left and right duals uses τ.

3.3 Tensor algebras and monomial relations

Example 3.6. *Quadratic duality exchanges having no relations with setting every product of two generators equal to zero. Taking $R = 0$ or $R = V \otimes_S V$ in (3.2) gives*

$$\big(T_S(V)\big)^! = T_S(V^*)/(V^* \otimes_S V^*) \cong S \oplus V^*,$$
$$\big(T_S(V)/(V \otimes_S V)\big)^! = T_S(V^*).$$

In $S \oplus V^*$, $(V^*)^2 = 0$ and $(s, f)(t, g) = (st, sg + ft)$.

For the rest of this subsection, take $S = \bigoplus_{i=1}^r \Bbbk e_i$, with $e_i e_j = \delta_{ij} e_i$ and $\sum_i e_i = 1$. Choose a basis $\mathcal{B}_{ij}$ of each $e_i V e_j$ and let $\mathcal{B} = \bigcup_{i,j} \mathcal{B}_{ij}$. A non-empty word $b_1 \cdots b_d$ is composable if $b_t \in \mathcal{B}_{i_{t-1}, i_t}$ for $1 \le t \le d$ and some $i_0, \ldots, i_d \in \{1, \ldots, r\}$. Thus adjacent factors have matching local units. By Lemma 2.14, the composable words form a basis of the positive degree part of $T_S(V)$.

Definition 3.7. *The presentation $A = T_S(V)/(R)$ is monomial quadratic with respect to $\mathcal{B}$ if R is spanned by a subset $\mathcal{R}$ of the composable words of length 2 in $\mathcal{B}$.*

For $b \in \mathcal{B}_{ij}$, define the right S-linear map $b^* : V \to S$ by its values on $\mathcal{B}$:

$$b^*(c) = \delta_{bc} e_j \quad (c \in \mathcal{B}), \qquad e_j b^* e_i = b^*. \tag{3.5}$$

The elements b^* form a $\Bbbk$-basis of V^*; the left and right local units of each basis element are reversed.

Proposition 3.8. *For a monomial quadratic presentation:*

(1) *A basis of A consists of the e_i and the non-empty composable words containing no adjacent pair from $\mathcal{R}$.*

(2) *The quadratic dual is monomial, with*

$$R^\perp = \operatorname{span}_\Bbbk\{c^* \otimes_S b^* \mid bc \text{ is composable and } bc \notin \mathcal{R}\}. \tag{3.6}$$

Proof The ideal generated by $\mathcal{R}$ is spanned by the composable words containing an element of $\mathcal{R}$ as an adjacent pair. Removing those basis words proves (1). For $b \in \mathcal{B}_{ij}$ and $c \in \mathcal{B}_{j\ell}$, (3.1) and (3.5) give

$$\theta_r(c^* \otimes b^*)(b \otimes c) = c^*(e_j c) = e_\ell.$$

The functional vanishes on every other basis tensor. By Lemma 2.14, the composable $c^* \otimes b^*$ form a basis of $V^* \otimes_S V^*$. Evaluating a linear combination on each $b \otimes c \in \mathcal{R}$ forces precisely the coefficient of $c^* \otimes b^*$ to be zero, since $e_\ell \neq 0$. Thus the annihilator is the span of the complementary basis tensors, proving (2). □

Example 3.9. *We apply Example 3.6 to the tensor algebra description of $\Bbbk A_n$ in Example 2.21. Take $A = \Bbbk A_n$, $S = \bigoplus_{i=1}^n \Bbbk e_{ii}$, and $V = \bigoplus_{i=1}^{n-1} \Bbbk e_{i,i+1}$. There are no quadratic relations, so $A^!$ has the reversed generators $e^*_{i,i+1} \in e_{i+1,i+1} V^* e_{ii}$ and every composable product of two generators is zero. With C as in Example 2.33, this gives*

$$H_{A^!}(t) = I_n + tC^{\mathsf{T}}.$$

Here ^{T} denotes transpose: $(C^{\mathsf{T}})_{ij} = C_{ji}$. For $n = 3$,

$$(\Bbbk A_3)^! \cong \big(\Bbbk A_3/(e_{13})\big)^{\mathrm{op}}, \qquad \big(\Bbbk A_3/(e_{13})\big)^! \cong (\Bbbk A_3)^{\mathrm{op}}.$$

These are graded isomorphisms fixing the diagonal idempotents.

Example 3.10. *For $A = \Bbbk A_2$, Examples 2.33 and 3.9 give*

$$H_A(t) = \begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix}, \qquad H_{A^!}(-t)^{\mathsf{T}} = \begin{pmatrix} 1 & -t \\ 0 & 1 \end{pmatrix}, \qquad H_A(t) H_{A^!}(-t)^{\mathsf{T}} = I_2.$$

The transpose aligns the indices because an S-dual reverses the two idempotents in (3.5).

Example 3.11. *Proposition 3.8 exchanges the forbidden and surviving composable pairs. Take* $A = T_S(V)/(ab)$ *from Example 2.23, where* $S = \Bbbk e_1 \oplus \Bbbk e_2$, $e_1 a e_2 = a$, *and* $e_2 b e_1 = b$. *The only composable words of length* 2 *are* ab *and* ba. *Thus*

$$A^! = T_S(V^*)/(a^*b^*), \qquad A^! = S \oplus \Bbbk a^* \oplus \Bbbk b^* \oplus \Bbbk b^* a^*.$$

The original relation $ab = 0$ *corresponds to the surviving word* b^*a^*; *the surviving word* ba *corresponds to the dual relation* $a^*b^* = 0$. *In the order* e_1, e_2, *their Hilbert series are*

$$H_A(t) = \begin{pmatrix} 1 & t \\ t & 1+t^2 \end{pmatrix}, \qquad H_{A^!}(t) = \begin{pmatrix} 1+t^2 & t \\ t & 1 \end{pmatrix}.$$

The degree 2 *terms occur in different entries:* $ba \in e_2 A e_2$, *whereas* $b^*a^* \in e_1 A^! e_1$.

3.4 Symmetric and exterior algebras

Proposition 3.12. *For finite-dimensional* $\Bbbk$*-vector spaces* V, W,

$$\mathsf{S}(V)^! \cong \bigwedge D(V), \qquad \Big(\bigwedge W\Big)^! \cong \mathsf{S}(D(W)).$$

Proof For $\mathsf{S}(V)$, choose a basis $x_1, \dots, x_n$ of V, with dual basis $x_1^*, \dots, x_n^*$ of $D(V)$. Here $S = \Bbbk$, and the evaluation (3.1) reads

$$\theta_r(x_j^* \otimes x_i^*)(x_p \otimes x_q) = \delta_{ip}\delta_{jq}.$$

A tensor $\sum_{i,j=1}^n c_{ij} x_i^* \otimes x_j^*$ vanishes on each $x_p \otimes x_q - x_q \otimes x_p$ precisely when $c_{pq} = c_{qp}$ for $1 \le p < q \le n$. Thus the orthogonal complement of the symmetric relations is spanned by $(x_i^*)^2$ for $1 \le i \le n$ and $x_i^* x_j^* + x_j^* x_i^*$ for $1 \le i < j \le n$, which are the exterior relations on $D(V)$.

For $\bigwedge W$, choose a basis $y_1, \dots, y_m$ of W, with dual basis $y_1^*, \dots, y_m^*$ of $D(W)$. A tensor $\sum_{i,j=1}^m c_{ij} y_i^* \otimes y_j^*$ annihilates the exterior relations precisely when $c_{ii} = 0$ for $1 \le i \le m$ and $c_{pq} + c_{qp} = 0$ for $1 \le p < q \le m$. The orthogonal complement is therefore spanned by $y_i^* y_j^* - y_j^* y_i^*$ for $1 \le i < j \le m$, which are the symmetric relations on $D(W)$. The presentations in (2.10) give both isomorphisms. □

Consequently, to pair $\mathsf{S}(V)$ with $\bigwedge W$ by quadratic duality, take $W = D(V)$ and $y_i = x_i^*$, so $m = n$. Evaluation $D(D(V)) \cong V$ from Example 1.10 then identifies $(\bigwedge W)^!$ with $\mathsf{S}(V)$.

The proof works in characteristic 2: the relations $(x_i^*)^2 = 0$ remain part of the exterior presentation even when the signs $+$ and $-$ coincide. Example 2.40 is the calculation for two generators. Quadratic duality need not preserve Hilbert series: if $\dim_\Bbbk V > 0$, Example 2.35 gives different series for $\mathsf{S}(V)$ and $\mathsf{S}(V)^! \cong \bigwedge D(V)$.

The *Koszul condition*, to be defined in Lecture 5, implies the identity

$$H_A(t) H_{A^!}(-t)^{\mathsf{T}} = I_r \qquad \text{when } A_0 = \bigoplus_{i=1}^r \Bbbk e_i. \tag{3.7}$$

This is [BGS96, Lemma 2.11.1]; the proof will be given in Lecture 8. Equation (3.7) is not asserted for arbitrary quadratic algebras. For $\mathsf{S}(V)$ and its quadratic dual $\bigwedge D(V)$, with $n = \dim_\Bbbk V$, the case $S = \Bbbk$ gives $(1-t)^{-n}(1-t)^n = 1$.

3.5 Direct sums

For graded algebras A, B, the finite direct sum $A \oplus B$ has componentwise multiplication, unit $(1_A, 1_B)$, and $(A \oplus B)_i = A_i \oplus B_i$. In particular, its degree 0 algebra is $A_0 \oplus B_0$.

Proposition 3.13. *Let $A = T_S(V)/(R)$ and $B = T_T(W)/(Q)$ be quadratic algebras. Then $A \oplus B$ is quadratic over $S \oplus T$, and*

$$(A \oplus B)^! \cong A^! \oplus B^!.$$

Proof Let $(s,t) \in S \oplus T$ act on V through s and on W through t, on both sides. The local unit $(1_S, 0)$ is the identity on V and acts as zero on W, so

$$v \otimes_{S\oplus T} w = v(1_S,0) \otimes_{S\oplus T} w = v \otimes_{S\oplus T} (1_S,0)w = 0 \quad (v \in V,\ w \in W).$$

Likewise $w \otimes_{S\oplus T} v = 0$. Distributing tensor products over direct sums by Lemma 2.2(4) gives

$$T_{S\oplus T}(V \oplus W) \cong T_S(V) \oplus T_T(W),$$
$$(V \oplus W) \otimes_{S\oplus T} (V \oplus W) \cong (V \otimes_S V) \oplus (W \otimes_T W).$$

The relation space is $R \oplus Q$. A right $(S \oplus T)$-linear map $V \oplus W \to S \oplus T$ maps V into S and W into T, so the generator dual is $V^* \oplus W^*$. The annihilator of $R \oplus Q$ is therefore $R^\perp \oplus Q^\perp$, giving the assertion by Definition 3.1. □

For example, $\big((\Bbbk A_2)^{\oplus 2}\big)^! \cong \big((\Bbbk A_2)^{\mathrm{op}}\big)^{\oplus 2}$ by Example 3.9. Both degree 0 algebras are $\Bbbk^{\oplus 4}$; there are no products connecting the two summands.

3.6 Tensor products and signs

Definition 3.14. *For graded $\Bbbk$-algebras A, B, the ordinary tensor product $A \otimes_\Bbbk B$ and the graded tensor product $A \otimes_\Bbbk^{\mathrm{gr}} B$ have underlying vector space $A \otimes_\Bbbk B$ and the total grading of Definition 2.30, with products*

$$\begin{aligned} (a \otimes b)(a' \otimes b') &= aa' \otimes bb' && \text{in } A \otimes_\Bbbk B, \\ (a \otimes b)(a' \otimes b') &= (-1)^{\deg b \deg a'} aa' \otimes bb' && \text{in } A \otimes_\Bbbk^{\mathrm{gr}} B \end{aligned} \tag{3.8}$$

for homogeneous $a, a' \in A$ and $b, b' \in B$, extended bilinearly.

Both products have identity $1_A \otimes 1_B$. For the graded product, associativity follows from $\deg b \deg a' + (\deg b + \deg b') \deg a'' = \deg b' \deg a'' + \deg b(\deg a' + \deg a'')$ for homogeneous $a', a'' \in A$ and $b, b' \in B$. For homogeneous $a \in A$ and $b \in B$, the elements $a \otimes 1$ and $1 \otimes b$ commute in the ordinary tensor product; in the graded tensor product they satisfy

$$(1 \otimes b)(a \otimes 1) = (-1)^{\deg a \ \deg b}(a \otimes 1)(1 \otimes b).$$

Suppose $A_0 = S$ and $B_0 = T$ are finite-dimensional semisimple and $S \otimes_\Bbbk T$ is semisimple. The sign does not change the underlying graded $(S \otimes_\Bbbk T)$-bimodule, so the two tensor products have the same Hilbert series whenever A, B are locally finite. For $S = \Bbbk^{\oplus r}$ and $T = \Bbbk^{\oplus s}$, Lemma 2.36 gives its entries:

$$H_{A\otimes_\Bbbk B}(t)_{(i,p),(j,q)} = H_A(t)_{ij} H_B(t)_{pq}.$$

In particular, when $S = T = \Bbbk$, this is $H_A(t)H_B(t)$.

Proposition 3.15. *Let $A = T_S(V)/(R)$ and $B = T_T(W)/(Q)$ be quadratic algebras, and assume that $S \otimes_\Bbbk T$ is semisimple. Then both tensor products are quadratic over $S \otimes_\Bbbk T$, and*

$$(A \otimes_\Bbbk B)^! \cong A^! \otimes_\Bbbk^{\mathrm{gr}} B^!, \qquad (A \otimes_\Bbbk^{\mathrm{gr}} B)^! \cong A^! \otimes_\Bbbk B^!. \tag{3.9}$$

The mixed commuting relations become anticommuting relations under the reversed tensor pairing.

The assumption on $S \otimes_{\Bbbk} T$ holds when S, T are products of matrix algebras over $\Bbbk$, since $\mathrm{Mat}_a(\Bbbk) \otimes_{\Bbbk} \mathrm{Mat}_b(\Bbbk) \cong \mathrm{Mat}_{ab}(\Bbbk)$. It does not follow from semisimplicity alone: take $S = T$ to be the field in Example 1.44.

Proof The degree 1 part of $A \otimes_{\Bbbk} B$ is $(V \otimes_{\Bbbk} T) \oplus (S \otimes_{\Bbbk} W)$. By Lemma 2.2, its tensor square over $S \otimes_{\Bbbk} T$ decomposes into the four summands

$$(V \otimes_S V) \otimes_{\Bbbk} T \;\oplus\; \underbrace{V \otimes_{\Bbbk} W}_{\text{order } VW} \;\oplus\; \underbrace{V \otimes_{\Bbbk} W}_{\text{order } WV} \;\oplus\; S \otimes_{\Bbbk} (W \otimes_T W).$$

Its quadratic relations consist of $R \otimes_{\Bbbk} T$, $S \otimes_{\Bbbk} Q$, and the mixed relations

$$(v \otimes 1)(1 \otimes w) - (1 \otimes w)(v \otimes 1) \quad (v \in V,\ w \in W). \tag{3.10}$$

Here the products are taken in the tensor algebra on the degree 1 part, and their span is an $(S \otimes_{\Bbbk} T)$-sub-bimodule. Let C be the quotient of that tensor algebra by the three specified relation spaces. The generator maps induce algebra homomorphisms $A \to C$, $a \mapsto \overline{a}$, and $B \to C$, $b \mapsto \overline{b}$. Their images commute: the degree 0 actions already commute across the two factors, and (3.10) makes the degree 1 generators commute. Consequently $A \otimes_{\Bbbk} B \to C$, $a \otimes b \mapsto \overline{a}\,\overline{b}$, is an algebra homomorphism. It is inverse to the canonical homomorphism $C \to A \otimes_{\Bbbk} B$, since both composites fix the generating copies of S, T, V, W. Thus the stated relations give exactly $A \otimes_{\Bbbk} B$. For the graded tensor product, the same construction uses $\overline{b}\,\overline{a} = (-1)^{\deg a \deg b} \overline{a}\,\overline{b}$ for homogeneous a, b.

The right $(S \otimes_{\Bbbk} T)$-duals of the two generator summands are $V^* \otimes_{\Bbbk} T$ and $S \otimes_{\Bbbk} W^*$, with evaluations

$$(f \otimes t)(v \otimes t') = f(v) \otimes tt', \qquad (s \otimes g)(s' \otimes w) = ss' \otimes g(w).$$

These identifications hold for finite free right modules and hence for their direct summands by Corollary 1.13. The reversed tensor evaluation pairs each pure summand with its corresponding dual summand and exchanges the two mixed summands; all other pairings are zero. The pure relation spaces have orthogonal complements $R^{\perp} \otimes_{\Bbbk} T$ and $S \otimes_{\Bbbk} Q^{\perp}$ in the corresponding pure summands. The mixed relation space is the set of pairs $(z, -z)$ in the two copies of $V \otimes_{\Bbbk} W$. For mixed tensors, the two reversed evaluations of $f \in V^*$ and $g \in W^*$ both give $f(v) \otimes g(w)$ on the corresponding copy. Thus an annihilating pair of functionals must have equal components. As these simple tensors span the full dual, the mixed annihilator is spanned by

$$(f \otimes 1)(1 \otimes g) + (1 \otimes g)(f \otimes 1) \quad (f \in V^*,\ g \in W^*).$$

These are the mixed relations of $A^! \otimes^{\mathrm{gr}}_{\Bbbk} B^!$, proving the first isomorphism in (3.9). For $A \otimes^{\mathrm{gr}}_{\Bbbk} B$, the minus sign in (3.10) is a plus sign. The mixed annihilator then has a minus sign, giving the ordinary tensor product on the dual side. □

Example 3.16. *We compute how Proposition 3.15 changes a commutation relation in a matrix example. Take $A = \Bbbk A_2 \otimes_{\Bbbk} \Bbbk A_2$. Its degree 0 algebra has basis $e_{ii} \otimes e_{jj}$ for $i, j \in \{1, 2\}$, and its degree 1 part has basis*

$$a = e_{12} \otimes e_{11}, \quad b = e_{22} \otimes e_{12}, \quad c = e_{11} \otimes e_{12}, \quad d = e_{12} \otimes e_{22}.$$

By Lemma 2.14, the tensor square of the degree 1 part has basis ab, cd, and both map to $e_{12} \otimes e_{12}$. Thus the relation space is $\Bbbk(ab - cd)$. Using the reversed generators from (3.5), its orthogonal complement is $\Bbbk(b^*a^* + d^*c^*)$. Hence the dual presentation has an anticommuting relation:

$$ab = cd \text{ in } A, \qquad b^*a^* = -d^*c^* \text{ in } A^!.$$

Order the degree 0 *idempotents as* $e_{11}\otimes e_{11}, e_{22}\otimes e_{11}, e_{11}\otimes e_{22}, e_{22}\otimes e_{22}$. *The homogeneous bases give*

$$H_A(t)=\begin{pmatrix}1 & t & t & t^2\\ 0 & 1 & 0 & t\\ 0 & 0 & 1 & t\\ 0 & 0 & 0 & 1\end{pmatrix}, \qquad H_{A^!}(t)=H_A(t)^{\mathsf{T}}.$$

The signs agree with Proposition 3.15; in characteristic 2, *the same presentations remain valid with* $-1=1$. *Here replacing* d^* *by* $-d^*$ *removes the minus sign. In this example the sign distinguishes the specified presentations, not the isomorphism classes of algebras.*

Example 3.17. *The symmetric/exterior pair shows that the sign in a graded tensor product cannot always be omitted. Let* $V=\Bbbk x_1\oplus\Bbbk x_2$ *and let* y_1, y_2 *be the dual basis of* $W=D(V)$. *Proposition 3.15 gives*

$$\mathsf{S}(V)^! \cong \big(\Bbbk[x_1]\otimes_{\Bbbk}\Bbbk[x_2]\big)^! \cong \frac{\Bbbk[y_1]}{(y_1^2)}\otimes_{\Bbbk}^{\mathrm{gr}}\frac{\Bbbk[y_2]}{(y_2^2)}\cong\bigwedge W.$$

If $\operatorname{char}\Bbbk\neq 2$, *the ordinary tensor product of the two square-zero algebras is commutative, whereas the graded tensor product is not:* $y_1y_2=-y_2y_1\neq 0$. *They are therefore not isomorphic. This sign cannot be removed by a change of generators.*

3.7 Exercises

Exercise 3.18. *For* $A=\Bbbk A_4/(e_{13})$, *write a presentation of* $A^!$ *using reversed generators and find homogeneous bases of* A *and* $A^!$. *Check the double dual using Proposition 3.8.* Hint. *Among* $e_{12}e_{23}$ *and* $e_{23}e_{34}$, *only the first is a relation. Complement the forbidden pairs and reverse their order.*

Exercise 3.19. *Let* $S=\Bbbk e_1\oplus\Bbbk e_2$ *and let* V *have basis* a, b *with* $e_1ae_2=a$ *and* $e_2be_1=b$. *Compute the quadratic dual of* $T_S(V)/(ab, ba)$ *and the Hilbert series of both algebras.* Check. *In the order* e_1, e_2, *the two series are*

$$\begin{pmatrix}1 & t\\ t & 1\end{pmatrix} \qquad \textit{and} \qquad \frac{1}{1-t^2}\begin{pmatrix}1 & t\\ t & 1\end{pmatrix}.$$

Exercise 3.20. *For the four generators in Example 3.16, replace the relation* $ab-cd$ *by* $ab-\lambda cd$, *where* $\lambda\in\Bbbk$. *Compute the quadratic dual relation for every* λ, *including* $\lambda=0$, *and recover the original relation by dualising again.* Hint. *Evaluate* $\mu b^*a^*+\nu d^*c^*$ *on* $ab-\lambda cd$. *For* $\lambda\neq 0$, *rescale* d *to compare with Example 3.16.*

4 Minimal graded resolutions

4.1 Projective covers and the graded Nakayama lemma

Let $A = \bigoplus_{d \geq 0} A_d$ be a locally finite graded algebra with $S = A_0$ finite-dimensional semisimple. Write

$$A_+ := \bigoplus_{d>0} A_d.$$

The grading makes A_+ a two-sided ideal, and projection onto degree 0 identifies A/A_+ with S. We regard S as a graded right A-module concentrated in degree 0, with action $s \cdot a = sa_0$ for $s \in S$ and $a = \sum_{d \geq 0} a_d \in A$.

All complexes below are chain complexes of graded right A-modules: their terms P_i are indexed by homological degrees $i \geq 0$, and their differentials are degree preserving maps $d_i : P_i \to P_{i-1}$ satisfying $d_{i-1}d_i = 0$ for $i \geq 2$. An augmented complex to a module M also has a degree preserving map $p_0 : P_0 \to M$ with $p_0 d_1 = 0$. The augmented complex is exact when $\operatorname{Im} d_{i+1} = \operatorname{Ker} d_i$ for $i \geq 1$, $\operatorname{Im} d_1 = \operatorname{Ker} p_0$, and p_0 is surjective.

For a graded right A-module M, define its *graded top* by

$$\overline{M} := M/MA_+.$$

The quotient $\overline{M}$ is a graded right S-module. We call M *bounded below* if $M_d = 0$ for all sufficiently small d. Every finitely generated graded module is bounded below: a finite generating set can be replaced by its finitely many homogeneous components, and A has no negative degrees.

Definition 4.1. *A graded right A-module P is projective if, for every surjective degree preserving map $q : X \to Y$ and every degree preserving map $f : P \to Y$, there is a degree preserving map $\widetilde{f} : P \to X$ such that $q\widetilde{f} = f$.*

The shifted free module $A(-d)$ has its generator 1 in degree d. If Q is a finite-dimensional projective right S-module, then $Q \otimes_S A(-d)$ is a projective graded right A-module: Q is a direct summand of a finite free right S-module, so $Q \otimes_S A(-d)$ is a direct summand of a finite direct sum of copies of $A(-d)$.

Lemma 4.2 (Graded Nakayama lemma). *If a bounded below graded right A-module M satisfies $M = MA_+$, then $M = 0$.*

Proof If $M \neq 0$, let d be the least degree with $M_d \neq 0$. Every degree d element of MA_+ is a sum of products $m_j a_{d-j}$ with $j < d$ and $d - j > 0$. All such m_j vanish by the choice of d, so $(MA_+)_d = 0$, contradicting $M_d = (MA_+)_d$. □

Definition 4.3. *A graded projective cover of a bounded below locally finite graded right A-module M is a surjective degree preserving map $p : P \to M$ from a bounded below locally finite graded projective module such that*

$$\overline{p} : P/PA_+ \xrightarrow{\sim} M/MA_+.$$

For a surjection $p : P \to M$, the condition in Definition 4.3 is equivalent to $\operatorname{Ker} p \subseteq PA_+$, since $p(PA_+) = MA_+$.

Proposition 4.4. *Every bounded below locally finite graded right A-module has a graded projective cover, unique up to isomorphism over M.*

Proof Put $\overline{M}_d = (M/MA_+)_d$. Since S is semisimple, the quotient map $M_d \to \overline{M}_d$ has a right S-linear section $s_d : \overline{M}_d \to M_d$. Set

$$P := \bigoplus_{d\in\mathbb{Z}} (\overline{M}_d \otimes_S A)(-d).$$

Each $\overline{M}_d$ is projective over S, so P is a graded projective right A-module. Since M is bounded below and locally finite, each degree of P is a finite sum of finite-dimensional spaces; hence P is locally finite. The maps

$$\overline{M}_d \otimes_S A \longrightarrow M, \qquad \overline{m} \otimes a \longmapsto s_d(\overline{m})a$$

define a degree preserving map $p : P \to M$. On graded tops, $\overline{p}$ identifies each copy of $\overline{M}_d$ with $(M/MA_+)_d$. Thus $\overline{p}$ is an isomorphism. The quotient $C := M/p(P)$ satisfies $C/CA_+ = 0$, hence $C = CA_+$; Lemma 4.2 gives $C = 0$. Therefore p is a projective cover.

For uniqueness, let $p : P \to M$ and $q : Q \to M$ be projective covers. Projectivity gives a map $f : P \to Q$ with $qf = p$. The maps $\overline{p}$ and $\overline{q}$ are isomorphisms, so $\overline{f} = \overline{q}^{-1}\overline{p}$ is an isomorphism. The quotient $Q/f(P)$ has zero graded top and is bounded below, hence is zero by Lemma 4.2. Thus f is surjective; since Q is projective, f splits. Its kernel is a direct summand of P, and its graded top is zero because $\overline{f}$ is injective. Lemma 4.2 gives $\operatorname{Ker} f = 0$, so f is an isomorphism over M. □

4.2 Minimal resolutions and graded Betti numbers

Definition 4.5. *A graded projective resolution of M is an exact sequence of degree preserving maps*

$$\cdots \longrightarrow P_2 \xrightarrow{d_2} P_1 \xrightarrow{d_1} P_0 \xrightarrow{p_0} M \longrightarrow 0$$

with every P_i graded projective. It is minimal if $p_0 : P_0 \to M$ is a graded projective cover and, for each $i \geq 1$, the map d_i with codomain restricted to $\operatorname{Ker} d_{i-1}$ is a graded projective cover, where $d_0 := p_0$.

Theorem 4.6. *Every bounded below locally finite graded right A-module has a minimal graded projective resolution. For any two such resolutions $P_\bullet \to M$ and $P'_\bullet \to M$, there are degree preserving isomorphisms $P_i \to P'_i$ commuting with the differentials and the maps to M.*

Proof Take a projective cover $p_0 : P_0 \to M$ using Proposition 4.4, and set $K_1 = \operatorname{Ker} p_0$. The module K_1 is bounded below and locally finite, so it has a projective cover $q_1 : P_1 \to K_1$. Repeat: after defining $q_i : P_i \to K_i$, set $K_{i+1} = \operatorname{Ker} q_i$ and cover K_{i+1} by $q_{i+1} : P_{i+1} \to K_{i+1}$. Composing each q_i with the inclusion $K_i \hookrightarrow P_{i-1}$ gives d_i and an exact minimal resolution.

For uniqueness, Proposition 4.4 identifies the covers of M in two minimal resolutions. The resulting isomorphism identifies their kernels; applying the same proposition to those kernels identifies the next projective covers. Induction gives an isomorphism in every homological degree, compatible with the differentials and the maps to M. □

Choose representatives $L_1, \ldots, L_r$ of the simple right S-modules, as in Definition 2.26. For a minimal resolution $P_\bullet \to M$, define the graded Betti numbers $\beta_{i,d,j}(M)$ by

$$(P_i/P_iA_+)_d \cong \bigoplus_{j=1}^{r} L_j^{\oplus \beta_{i,d,j}(M)}, \qquad \beta_{i,d}(M) := (\beta_{i,d,1}(M), \ldots, \beta_{i,d,r}(M)).$$

Here i is the homological degree, d is the grading degree, and j records the simple right S-module. Theorem 4.6 shows these multiplicities do not depend on the chosen minimal resolution. The projective cover construction also gives

$$P_i \cong \bigoplus_{d\in\mathbb{Z}} \bigoplus_{j=1}^{r} (L_j \otimes_S A)(-d)^{\oplus \beta_{i,d,j}(M)}.$$

For $S = \bigoplus_{j=1}^{r} \Bbbk e_j$, the simple modules are $L_j = e_j S$ and $L_j \otimes_S A \cong e_j A$.

Example 4.7. *This example shows how a non-field degree zero algebra gives distinct projective covers for its different simple modules. Let $A = \Bbbk A_2$, with $S = \Bbbk e_{11} \oplus \Bbbk e_{22}$ and $A_1 = \Bbbk e_{12}$. The simple right S-modules are $L_1 = e_{11}S$ and $L_2 = e_{22}S$, with projective lifts $Q_1 = e_{11}A$ and $Q_2 = e_{22}A$. The quotient map $Q_1 \to L_1$ has kernel $\Bbbk e_{12}$, while $Q_2 = L_2$. Sending the degree 1 generator e_{22} to e_{12} gives the minimal resolution*

$$0 \longrightarrow Q_2(-1) \xrightarrow{d_1} Q_1 \longrightarrow L_1 \longrightarrow 0, \qquad d_1(e_{22}) = e_{12}.$$

Example 4.8. *For $A = \Bbbk[x]$, with $\deg x = 1$, the degree zero quotient module is $\Bbbk = A/(x)$. Multiplication by x gives the minimal resolution*

$$0 \longrightarrow A(-1) \xrightarrow{d_1} A \xrightarrow{\pi} \Bbbk \longrightarrow 0, \qquad d_1(a) = xa, \quad \pi(f) = f(0).$$

The map d_1 has degree 0, is injective, and has image $xA = \operatorname{Ker} \pi \subset AA_+$. Thus $\beta_{0,0} = 1$ and $\beta_{1,1} = 1$.

Example 4.9. *The exterior algebra on one generator gives an infinite minimal resolution, although the algebra itself is finite-dimensional. Let $E = \Bbbk[\epsilon]/(\epsilon^2)$ with $\deg \epsilon = 1$ and $E_+ = \Bbbk\epsilon$. The degree zero quotient is $\Bbbk = E/E_+$, and*

$$\cdots \xrightarrow{d_3} E(-2) \xrightarrow{d_2} E(-1) \xrightarrow{d_1} E \xrightarrow{\pi} \Bbbk \longrightarrow 0, \qquad d_i(a) = \epsilon a, \quad \pi(a) = a_0.$$

Each d_i is degree preserving and right E-linear. Since $\operatorname{Ker}(\epsilon \cdot -) = \epsilon E = \operatorname{Im}(\epsilon \cdot -)$, the sequence is exact at every $E(-i)$ and at E. Every differential has image in the preceding term times E_+, so the resolution is minimal and $\beta_{i,i} = 1$ for all $i \geq 0$.

If M is finitely generated, its first projective cover is finitely generated. Without a noetherian hypothesis on A, later kernels need not be finitely generated, so later terms of the minimal resolution need not be finitely generated. The bounded below locally finite setting above still gives projective covers and a minimal resolution.

5 Koszul complex and Koszul algebras

5.1 The Koszul complex of a polynomial algebra

Let V be an n-dimensional $\Bbbk$-vector space, choose a basis $x_1, \dots, x_n$, and let $A = \mathsf{S}(V) = \Bbbk[x_1, \dots, x_n]$, with each x_j in degree 1. Let $\pi : A \to \Bbbk$ be projection onto degree 0, so $\Bbbk = A/A_+$. For $0 \le i \le n$, put

$$K_i := \bigwedge^i V \otimes_{\Bbbk} A(-i), \qquad K_0 = A.$$

The right A-action is on the second factor. The factor $\bigwedge^i V$ is placed in grading degree 0, so the generators of K_i have grading degree i.

Definition 5.1. *For $1 \le i \le n$, define the Koszul differential*

$$\begin{aligned} d_i : K_i &\longrightarrow K_{i-1}, \\ (v_1 \wedge \cdots \wedge v_i) \otimes a &\longmapsto \sum_{r=1}^{i} (-1)^{r-1} (v_1 \wedge \cdots \wedge \widehat{v_r} \wedge \cdots \wedge v_i) \otimes v_r a. \end{aligned} \tag{5.1}$$

The formula is multilinear and alternating in $v_1, \dots, v_i$, so it defines a map on $\bigwedge^i V$. The augmented chain complex $K_\bullet \to \Bbbk$ consists of these maps and $\pi : K_0 = A \to \Bbbk$.

Proposition 5.2. *The augmented complex*

$$0 \longrightarrow K_n \xrightarrow{d_n} \cdots \xrightarrow{d_2} K_1 \xrightarrow{d_1} K_0 \xrightarrow{\pi} \Bbbk \longrightarrow 0$$

is an exact minimal graded free resolution of $\Bbbk$ over A.

Proof Each d_i preserves degree and is right A-linear. For $i \ge 2$, compute $d_{i-1}d_i$ on a basis wedge. Every term is indexed by two distinct positions $r < s$ deleted from that wedge. Deleting position r and then s gives coefficient $(-1)^{r+s-3} x_{j_s} x_{j_r}$; deleting s and then r gives coefficient $(-1)^{r+s-2} x_{j_r} x_{j_s}$. Since $x_{j_s} x_{j_r} = x_{j_r} x_{j_s}$, these two terms sum to zero. Hence $d_{i-1}d_i = 0$; also $\pi d_1 = 0$ because every term of d_1 is divisible by a variable.

To prove exactness, give A and $\bigwedge V$ the multigrading with $\deg x_j$ equal to the jth standard basis vector. For $\alpha \in \mathbb{Z}^n_{\ge 0}$, let $\operatorname{supp}(\alpha) = \{j \mid \alpha_j > 0\}$. The degree α part of K_i has basis

$$b_J := (x_{j_1} \wedge \cdots \wedge x_{j_i}) \otimes x^{\alpha - \mathbf{1}_J}, \qquad J = \{j_1 < \cdots < j_i\} \subseteq \operatorname{supp}(\alpha),$$

where $\mathbf{1}_J$ has entry 1 at indices in J and 0 elsewhere. The differential is $d_i(b_J) = \sum_{r=1}^{i} (-1)^{r-1} b_{J \setminus \{j_r\}}$.

If $\alpha \ne 0$, let q be the least index in $\operatorname{supp}(\alpha)$. For each $i \ge 0$, define $h_{\alpha,i} : K_{i,\alpha} \to K_{i+1,\alpha}$ on basis vectors by

$$h_{\alpha,i}(b_J) = \begin{cases} b_{\{q\} \cup J}, & q \notin J, \\ 0, & q \in J, \end{cases}$$

inserting q first in the ordered wedge. For $q \notin J$ and $i \ge 1$, direct substitution gives

$$\begin{aligned} d_{i+1} h_{\alpha,i}(b_J) &= b_J - \sum_{r=1}^{i} (-1)^{r-1} b_{\{q\} \cup (J \setminus \{j_r\})}, \\ h_{\alpha,i-1} d_i(b_J) &= \sum_{r=1}^{i} (-1)^{r-1} b_{\{q\} \cup (J \setminus \{j_r\})}. \end{aligned}$$

For $i = 0$, $J = \varnothing$ and $d_1 h_{\alpha,0}(b_\varnothing) = b_\varnothing$. For $q \in J$, the term of $d_i(b_J)$ deleting q is the only term on which $h_{\alpha,i-1}$ is nonzero, and it gives $h_{\alpha,i-1} d_i(b_J) = b_J$ while $d_{i+1} h_{\alpha,i}(b_J) = 0$. Thus $d_{i+1} h_{\alpha,i} + h_{\alpha,i-1} d_i =$

$\mathrm{id}_{K_{i,\alpha}}$ for every $i \geq 1$ in positive multidegree; the case $i = 0$ was checked separately. All positive multidegrees are therefore exact. In multidegree 0, only $K_0 \cong \Bbbk$ is nonzero and π is the identity onto $\Bbbk$. This proves exactness of the augmented complex.

Finally, each K_i is a finite direct sum of copies of $A(-i)$, and (5.1) has image in $K_{i-1}A_+$. Exactness gives $\operatorname{Ker} d_i = \operatorname{Im} d_{i+1} \subseteq K_iA_+$ for $i < n$, and $\operatorname{Ker} d_n = 0$. Also, $\operatorname{Ker} \pi = AA_+$. Thus each map from K_i onto the preceding kernel has kernel contained in K_iA_+, so it is a projective cover by Definition 4.3. The resolution is therefore minimal. □

5.2 Koszul algebras

Definition 5.3. *A minimal graded projective resolution $P_\bullet \to M$ is linear if P_i/P_iA_+ is concentrated in degree i for every $i \geq 0$. This is equivalent to P_i being generated in degree i: generation in degree i implies the stated concentration because A_+ raises degree. Conversely, if the top is concentrated in degree i, boundedness below and Lemma 4.2 show that P_i has no components below degree i; the quotient $P_i/(P_i)_iA$ then has zero graded top, so Lemma 4.2 makes that quotient zero.*

Definition 5.4. *Let $A = T_S(V)/(R)$ be a quadratic algebra with $S = A_0$ finite-dimensional semisimple. The algebra A is Koszul if the degree zero quotient module $S = A/A_+$ has a linear minimal graded projective resolution as a right A-module.*

The generators in homological degree i must then all have grading degree i. Since S can have several simple modules, the definition requires a linear resolution of their direct sum, not just of one chosen simple module. For a minimal resolution, Definition 5.3 is equivalently the vanishing $\beta_{i,d,j}(S) = 0$ whenever $d \neq i$.

Proposition 5.5. *For every finite-dimensional vector space V, the symmetric algebra $\mathsf{S}(V)$ is Koszul.*

Proof Proposition 5.2 gives a minimal resolution with $K_i = \bigwedge^i V \otimes_\Bbbk A(-i)$. Thus K_i is generated in degree i for every i. □

Example 5.6. *The definition also applies when S has several simple modules. For $A = \Bbbk A_2$, the resolution in Example 4.7 is linear for L_1, and $L_2 = e_{22}A$ is projective in degree 0. Taking their direct sum gives a linear resolution of $S = L_1 \oplus L_2$; hence $\Bbbk A_2$ is Koszul.*

Example 5.7. *The exterior algebra $E = \Bbbk[\epsilon]/(\epsilon^2)$ is quadratic, and Example 4.9 gives a minimal resolution of its degree zero quotient with $P_i = E(-i)$. Thus E is Koszul, although its minimal resolution is infinite.*

List of Symbols

Page numbers link to the first defining passage. Entries marked advanced refer to optional material.

Symbol	Meaning	Page
Algebras and categories		
$\Bbbk$	Base field.	1
$\mathrm{Mat}_n(\Bbbk)$	Algebra of $n \times n$ matrices over $\Bbbk$.	2
$\Bbbk A_n$	Upper triangular $n \times n$ matrix algebra.	2
e_{ij}	Matrix unit with 1 in position (i, j).	2
$\mathrm{Hom}_{\mathcal{C}}(X, Y),\ \mathrm{id}_X$	Morphisms from X to Y; identity of X.	2
$\mathrm{Mod}\, A$	Right A-modules and A-linear maps.	2
$\mathrm{mod}\, A$	Finitely generated right A-modules.	2
$\mathrm{Hom}_A(M, N)$	A-linear maps, with no degree restriction.	2
$\mathrm{add}\, M$	Direct summands of finite direct sums of the right module M.	2
$\mathcal{C}^{\mathrm{op}}$	Opposite category: morphisms have their directions reversed.	3
$\mathrm{rad}\, A$	Jacobson radical of the algebra A.	4
$eA,\ eAe$	Right ideal and corner algebra associated with an idempotent e.	5
A^{op}	Opposite algebra: $a^{\mathrm{op}} b^{\mathrm{op}} = (ba)^{\mathrm{op}}$.	6
$\mathrm{Mod}(A^{\mathrm{op}})$	Left A-modules, viewed as right A^{op}-modules.	6
$\mathrm{add}_{\mathrm{mod}\, A^{\mathrm{op}}}(A)$	Direct summands of finite direct sums of the left regular module.	24
Gradings, sums, and bimodules		
A_g	Degree g component of a group-graded algebra.	6
M_g	Degree g component of a graded module.	6
$M(d)$	Grading shift: $M(d)_j = M_{j+d}$.	7
$\mathrm{Hom}_A^{\mathbb{Z}}(M, N)$	Degree-preserving A-linear maps.	7
$\mathrm{Hom}_A^{\mathbb{Z}}(M, N(d))$	A-linear maps $M \to N$ of degree d.	7
$\mathrm{GrMod}\, A$	Graded right A-modules and degree-preserving maps.	7
$\mathrm{grmod}\, A$	Locally finite graded right A-modules; finite generation is not required.	7
A_+	Positive degree ideal $\bigoplus_{i>0} A_i$.	9
$\pi : A \to A_0$	Projection onto degree 0, identifying A/A_+ with A_0.	9
$\bigoplus_{i \in I} M_i$	Direct sum, with finite support.	10
$M^{\oplus r}$	Direct sum of r copies of M.	2
$\prod_{i \in I} M_i$	Direct product, allowing arbitrary coordinates.	10
$\Bbbk[[x]]$	Formal power series ring.	10
${}_B M_C$	Bimodule with left B-action and right C-action.	11
A^{en}	Enveloping algebra $A \otimes_{\Bbbk} A^{\mathrm{op}}$.	12
$S = A_0$	Finite-dimensional semisimple degree zero algebra.	14
Duals and comparison maps		
$D(V)$	$\Bbbk$-linear dual $\mathrm{Hom}_{\Bbbk}(V, \Bbbk)$.	4
$D(u)$	Dual map, given by precomposition with u.	4
ev_V	Evaluation $V \to D(D(V))$, sending v to $\lambda \mapsto \lambda(v)$.	4
$\mathbb{D}(V)$	Graded $\Bbbk$-linear dual, with $\mathbb{D}(V)_i = D(V_{-i})$.	13

List of Symbols (continued)

Symbol	Meaning	Page
$M^*,\ {}^*N$	Right and left S-linear duals: $\mathrm{Hom}_S(M,S)$ and $\mathrm{Hom}_{S^{\mathrm{op}}}(N,S)$.	14
$M^{(*)},\ {}^{(*)}N$	Graded right and left S-duals, with degrees reversed.	14
$M^{*_{\mathrm{en}}}$	Two-sided dual $\mathrm{Hom}_{S^{\mathrm{en}}}(M,S^{\mathrm{en}})$; advanced.	16
$M^{(*_{\mathrm{en}})}$	Graded two-sided dual, with degrees reversed; advanced.	16
$\tau : S \to \Bbbk$	Symmetrising form on S.	17
$\alpha_M,\ \beta_N$	Comparisons from S-duals to linear duals, by composition with τ.	17
γ_M	Comparison from the two-sided dual to the right S-dual; advanced.	19
Tensor products, presentations, and Hilbert series		
$U \otimes_{\Bbbk} V$	Tensor product of vector spaces.	9
$M \otimes_A N$	Tensor product over A, with $ma \otimes n = m \otimes an$.	21
$V^{\otimes_A i}$	ith tensor power over A; $V^{\otimes_A 0} = A$.	24
$\theta_r,\ \theta_\ell$	Tensor-Hom comparison maps; tensor-dual isomorphisms in the semisimple case.	24
$\mu_i : A_1^{\otimes_S i} \to A_i$	Multiplication of i elements of degree 1.	26
$T_S(V)$	Tensor algebra generated by the bimodule ${}_SV_S$.	27
$\mathsf{S}(V)$	Symmetric algebra of V.	27
$\bigwedge W$	Exterior algebra of W.	27
$I = \mathrm{Ker}\,\mu$	Ideal of relations for $\mu : T_S(A_1) \to A$.	29
$A = T_S(V)/(R)$	Quadratic presentation: generators V in degree 1, relations R in degree 2.	29
$L_1, \dots, L_r$	Representatives of the simple right S-modules.	30
$\mathbf{dim}M$	Row vector of simple-module multiplicities in M_S.	30
$\mathbf{dim}U$	Bimodule dimension array; row i is $\mathbf{dim}(L_i \otimes_S U)$.	30
$H_M(t)$	Module Hilbert series $\sum_{d\in\mathbb{Z}}(\mathbf{dim}M_d)t^d$.	31
$H_A(t)$	Algebra Hilbert series, with entries recording both simple-module indices.	31
qA	Quadratic part $T_S(V)/(I_2)$ of $A = T_S(V)/I$; advanced.	33
$U^{\perp},\ {}^{\perp}U$	Right and left S-linear functionals vanishing on U.	34
Quadratic duality and resolutions		
$A^!,\ {}^!A$	Right and left quadratic dual algebras.	36
$A \otimes_{\Bbbk} B$	Ordinary algebra tensor product, without signs in multiplication.	40
$A \otimes_{\Bbbk}^{\mathrm{gr}} B$	Graded algebra tensor product, with the sign in (3.8).	40
$d_i : P_i \to P_{i-1}$	Chain differential; preserves grading degree and lowers homological degree.	43
$\overline{M}$	Graded top M/MA_+.	43
$P_\bullet \to M$	Graded projective resolution of M.	44
$\beta_{i,d,j}(M),\ \beta_{i,d}(M)$	Graded Betti multiplicities and their row vector; i is homological degree, d is grading degree, and j indexes the simple module.	44
$K_i,\ K_\bullet$	Koszul terms $K_i = \bigwedge^i V \otimes_{\Bbbk} A(-i)$ and their complex, for $A = \mathsf{S}(V)$.	46
$d_i : K_i \to K_{i-1}$	Koszul differential.	46

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